SAINT MICHAEL COLLEGE OF CARAGA

CENTRALIZED CLIENT INFORMATION MANAGEMENT SYSTEM FOR BUTUAN CITY SOCIAL WELFARE DEVELOPMENT DEPARTMENT

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CHAPTER 1

INTRODUCTION

1.1 Project Context

In today's democracies, social welfare is crucial for ensuring that individuals' fundamental needs are met and their quality of life improves. This study is significant for ending social isolation, reducing poverty, and ensuring that everyone has equal access to resources and opportunities. It is essential to utilize the latest digital tools to ensure that social assistance and welfare programs are accurate, accessible, and easily located. The Department of Social Welfare and Development (DSWD) in the Philippines faces a greater challenge in maintaining track of client records now that semi-digital tools like Google Sheets and Excel spreadsheets are available. It can manage a large amount of data, but it often encounters difficulties such as delays and duplicate data, which makes it harder for the agency to assist individuals quickly.

People all over the world are becoming increasingly aware of the challenges associated with handling data in traditional ways within the public aid sector. Deltek [1] stressed how hard it is to use spreadsheet-based systems to store private information, especially when many people are looking at the same data at the same time. As a result, clients' information is often at risk of being incorrect or stolen. Tzagkarakis [2] also stated that centralized systems enhance government services, openness, and accountability. He also claimed that efficient data management is directly related to the success of social activities. Many countries have found that their social services work more effectively and are easier to access after implementing these procedures.

The Philippines' Office of Senior Citizen Affairs (OSCA) utilizes an Electronic Records Management System (ERMS) to manage its records effectively. The Butuan City Social Welfare and Development Department (BCSWDD) still faces challenges, nevertheless, such as data fragmentation and delays in delivering services [3]. The Butuan City Social Welfare and Development Department (BCSWDD) faced similar challenges in maintaining accurate records of client information. The department now manages records using semi-digital methods, which makes things less efficient and slows down service delivery. Old and outdated technology makes it difficult to obtain accurate reports, collect data in real-time, and track customer development. When information is spread across several systems and departments, it is challenging to communicate effectively and collaborate. It does not help that there is no single, comprehensive database that encompasses all of this information. The department's social services are often less effective due to delays and inefficiencies, as there is no central organization.

A Centralized Client Information Management System (CCIMS) is proposed as a solution to these challenges. All client records from various departments of the Butuan City Social Welfare and Development Department will be consolidated into a single system. CCIMS will provide you with a centralized, web-based platform for managing customer data, ensuring that it is accurate, up-to-date, and readily available at all times. The solution will enhance service delivery, streamline case management, and minimize the likelihood of data encoding errors by centralizing data. Social workers and administrators will be able to make better decisions and utilize resources more efficiently if they can access client information more quickly and easily. The system will also create reports automatically, which will make it easier to keep an eye on the program and make services more open.

CCIMS is prompted by the Butuan City Social Welfare and Development Department's insufficient procedures and challenges in maintaining client records. The planned CCIMS aims to enhance social service delivery by reducing data fragmentation, minimizing errors in manual processes, and addressing other issues. CCIMS's use of a standardized data management system will enable the department to make better decisions more quickly and better meet the needs of the most disadvantaged groups in society. Furthermore, ensuring that client information is secure and accessible only to those in positions of authority would increase staff accountability. This research will inform the system's design and development, ensuring it meets the BCSWDD's needs and adheres to best practices in system design and implementation.

CCIMS is crucial because it could significantly improve the effectiveness and efficiency of the social services that the City of Butuan provides. CCIMS can help government agencies utilize their resources more effectively, reduce errors in their paperwork, and manage data more efficiently. On the other hand, social workers will have more time to help clients directly, rather than spending time on paperwork. Clients, especially those in high-risk locations, would benefit from services that are faster and more accurate, ensuring they receive the aid they need when they need it. Clients would be able to see more easily what services they are getting with one system. Finally, this study contributes to the growing body of research on digital transformation in the public sector by providing valuable insights for other local governments (LGUs) seeking to enhance their existing social programs. By making social services more effective, accessible, and open, CCIMS can help achieve bigger goals of fairness and social justice.

1. 2 Literature Review

Client Relationship Management (CRM) in Public Sector Client Information Systems

Client Relationship Management (CRM) is an important method that businesses use to manage clients and citizens. The CRM is very important in public sectors to centered the data, facilitate communication, and improve service provision to increase the people's satisfaction and strengthen transparency. This method is especially important for systems such as the Centralized Client Information Management System for Butuan City Social Welfare Development Department (CCIMS-BCSWDD), which relies on CRM's effective strategies to manage clients' information and improve the delivery of public services.

Numerous studies have demonstrated that integrating Client Relationship Management (CRM) systems can significantly enhance an organization's service and efficiency. For example, Sabahat Ansari et al. have reviewed[4] How Flutter utilizes a Client Management System helps facilitate client data management, enhance the user experience, and provide real-time updates. Similarly, D. Riananingrum et al. [5] demonstrate that centralized customer service systems can lose focus on citizen data, which hinders government agencies from providing better services and reducing wait times. Meanwhile, Nzube Promise Boniface et al. [6] have explored CRM methods in the vehicle industry. However, their views on adapted communication and customer satisfaction can be adapted to public sector organizations, especially in the management of relationships with citizens. Michael A. Ricci et al also emphasized. [7] The importance of client service by reviewing how CRM principles can enhance communication with the people and cause more successful public administration audits. Moreover, Neelesh et al. [8] present a centralized client connection system with CRM solutions, which shows how such systems facilitate client data management and increase transparency, which is particularly useful for public sector agencies that manage sensitive information. In total, these studies

show that CRM systems, when adapted to the public sector's needs, can improve the cooperation of agencies, quality of service, and citizen information management.

CRM principles can be used in the CCIMS-BCSWDD to generate a centralized system for more effective information management of beneficiaries. Additionally, it can be costly for small or constrained government agencies to develop and maintain CRM systems [7]. When handling personal information, it is essential to consider privacy issues and comply with data protection requirements [4].

Although CRM systems offer numerous benefits, several issues have been recognized, particularly in the implementation of public sector systems. Incorporating CRM systems into existing government databases presents a significant challenge, as it is a complex and resource-intensive process [5]. Moreover, it may be challenging for some public sector workers to adopt new technologies, which limits the full utilization of CRM systems [4]. Additionally, it can be costly to build and keep up CRM systems, especially for small or constrained government entities [7]. When dealing with people's private information, privacy concerns and the need to comply with data protection rules are also significant considerations [4].

Studies stated that the writing keeps talking about how using Client Relationship Management (CRM) tools can help the public sector provide better services and manage their clients better. Ansari et al. [4] and Ricci et al. [7] showed that using CRM systems improves managing client data and getting in touch with citizens, which makes public service delivery and government work better. This study

agrees with Boniface et al.'s results, which show how governments can use CRM strategies from other industries, like the auto industry, to make people happier and improve relations between the government and the public. Another thing that Neelesh et al. [8] CRM systems can enhance transparency, making it crucial for managing sensitive data within the government sector.

However, the difficulties in putting these ideas into practice are also covered in much research. Riananingrum et al. [5] state that it is difficult and time-consuming to integrate CRM systems with outdated and disjointed government datasets. According to Ansari et al.[4], this problem is exacerbated by the reluctance of some government employees to adopt new technologies, which underutilizes the full potential of CRM. Another issue raised is the high cost of establishing and maintaining these systems. This study presents a significant obstacle, particularly for small government organizations.

In summary, although studies indicate that CRM systems have significant potential for improving public services, challenges related to integration, costs, and user acceptance still need to be addressed. CCIMS-BCSWDD should be designed to respond to the specific needs of each division, ensure data security, and encourage staff acceptance. Despite the limitations, the overall literature review indicates that when well implemented, CRM principles can significantly contribute to enhancing client management, improving service efficiency, and promoting transparency in public institutions such as CCIMS-BCSWDD.

Security and Blockchain in Data Management Systems

As more individuals utilize digital information management systems, it is becoming increasingly evident that public sector applications require robust data security and blockchain technologies. It is crucial to ensure that data is safe, private, and accurate when using CCIMS, which handles sensitive client data for social welfare programs. The goals can be helped by blockchain acting as a decentralized ledger system. It is because it allows for precise, unchangeable, and secure record-keeping. Security protocols make sure that only authorized users can access the records. This theme looks at how blockchain technology and security measures can make data management systems like CCIMS better by providing both technical solutions and real-world uses for protecting data integrity and privacy.

Several studies demonstrate the importance of data security and blockchain technology in handling sensitive information. Duggineni [9] emphasizes the importance of ensuring data safety and accuracy in digital systems, particularly in government and healthcare settings. His study, which primarily focuses on data encryption, access control, and audit trails, aligns with CCIMS's objectives of maintaining client data security and accuracy. There is also a blockchain framework for managing healthcare data developed by Al-Khasawneh et al [10]. To keep data safe, they recommended using smart contracts and managing secret keys. This study examines the potential application of blockchain in CCIMS, offering an immutable record that safeguards data, maintains clarity, and restricts access to authorized individuals only.

On the other hand, Islam & Apu [11] compared centralized and decentralized blockchain solutions, showing that decentralized systems are safer but struggle with growth. A hybrid approach that combines the safety of autonomous blockchains with the ability to scale could be beneficial for CCIMS. Nguyen et al.[12] also discussed methods for protecting client privacy using Private Information Retrieval (PIR). These steps could help CCIMS prevent unauthorized individuals from

accessing private client data, making it safer and more secure. Dhanwate et al. [13] researched user management systems, demonstrating the importance of identification and role-based access control (RBAC) in ensuring that only authorized individuals can access specific types of client information. In the last part, Lisiak-Felicka and Szmit [14] looked at how local government offices deal with problems with information security. They emphasized the importance of monitoring things in real-time and acting quickly when issues arise. Similar incident management techniques, such as auditing and real-time reporting, may be employed by CCIMS to handle data leaks more efficiently and with reduced risk.

The findings underscore the importance of data security and blockchain in managing sensitive information across various sectors, including healthcare, public administration, and social welfare. Encryption, access control, and audit logs are the most crucial components of these systems, ensuring data remains safe and private. The adoption of blockchain for securing and auditing transactions also aligns with CCIMS, which could benefit from these technologies in ensuring transparency, preventing fraud, and securing client data against unauthorized access.

Although researchers equally emphasize the importance of data security and blockchain, there is a difference in the way of application. Blockchain technology provides strong protections such as data immutability and decentralized systems, but it has challenges when it comes to scalability or the ability to process large amounts of data, as mentioned by Islam and Apu [11] note that it is challenging to scale and process massive amounts of data. Meanwhile, centralized systems, although faster and more efficient, tend to have weaknesses such as a single point of failure, according to studies regarding PIR and user management. When blockchain and data security measures are integrated into CCIMS, they will enhance client information, making it more open, accountable, and secure.

Decentralized systems are generally safe, but they still have issues with speed and handling large amounts of data. Because of this, a hybrid method is considered a viable approach to solving the CCIMS problem. In addition, utilizing incident management and real-time warning systems will further enhance the system's safety by enabling the prompt identification and mitigation of threats. Through the combined use of encryption, blockchain, and proactive incident management, CCIMS can create a secure, reliable, and transparent system for managing clients' sensitive data.

In the total synthesis of the studies, it is clear that integrating blockchain technology and security measures into systems like CCIMS makes a significant contribution to maintaining the transparency, accountability, and privacy of information. Although the security of decentralized blockchain systems is more stable, they have a limitation in scalability; therefore, the use of hybrid models is more beneficial. The integration of real-time monitoring and incident alerts also enables the quick identification and prevention of any risk. By combining encryption, blockchain technology, and proactive incident management, CCIMS can create a secure, scalable, and transparent system that efficiently addresses both security and performance issues.

Digital Transformation in Public Sector Information System

Digital transformation in government offices means using technology to make them more efficient, better at providing services, and more open to the public. To keep up with the changing needs of the public, this transformation encompasses not only automating processes but also a significant shift in ways of doing things, cultures, and technology. Through the digitization of services, governments can provide more accurate, efficient, and easily accessible public services, as demonstrated in various studies

on the integration of technology in public processes.

Numerous studies have highlighted the growing importance of digital technology and workflow automation in enhancing government operations and public services. Reviewed by Malubag et al. [15] How workflow automation tools improve operational efficiency, and they discovered significant benefits to speed, accuracy, and collaboration with government agencies. Similarly, Syaidaturrahmi et al. [16] presented an integrated information system for health data reporting in Indonesia, demonstrating how web-based systems enhance data accuracy and reporting approaches that can be applied to the centralized design of CCIMS. Jonathan [17] reviewed digital transformation in the public sector, focusing on critical aspects of leadership, organizational culture, and technology in the successful adoption of digital solutions. Daud et al [18] deployed the Local Government Information System (SIPD) in Gorontalo, Indonesia. They demonstrated how digital financial systems enhance transparency and accountability, principles that align with the goals of CCIMS. In addition, Botangen et al.[3] made the web-based Records Management System for senior citizen data in the Philippines. This study makes data more accessible and improves working efficiency. It also shows how to keep client data on CCIMS. In support of these results, McKillop et al. [19] emphasize the need for handling different social welfare programs in a manner that is both flexible and scalable. Moon et al.[20] How data-driven methods in homeless services enhance resource utilization and service delivery by combining data. Studies on financial management, such as those by Owin et al. [21] and Juma et al. [22] have shown that combined systems, like IPPS and If, clarify processes, reduce fraud, and hold individuals more accountable for their finances. These are all goals of CCIMS. Finally, Adewumi and Abasilim [23] emphasized the importance of this for public sector workers. They said that ongoing training and skill development are key to the smooth operation and long-term success of digital systems like CCIMS.

Ultimately, Adewumi and Abasilim [23] emphasized the need for digital competency in public sector professionals, stating that continuous training and skill development are essential for the successful implementation and long-term success of digital systems, such as CCIMS. In total, studies have demonstrated that integrating digital technology, automation, and data management tools can lead to significant and positive improvements in efficiency, transparency, and accountability in public sector operations.

In total, studies have shown that the integration of digital technology, automation, and data management tools can cause significant and positive change in efficiency, transparency, and accountability to public sector operations.

These studies have demonstrated that the successful integration of digital systems in government agencies necessitates both human and technological infrastructure, as well as the requisite skills and expertise. Systems such as CCIMS can enhance the efficiency, transparency, and accessibility of government services by automating workflows, improving data accuracy, and facilitating better coordination among departments. However, studies also reveal challenges in implementation, including resistance to change, a lack of digital expertise, and the need for ongoing training and support from leaders to implement these systems successfully.

Studies show that going digital makes services more efficient and better delivered. However, Jonathan [17] and Adewumi and Abasilim [23] argue that implementing these systems in the public sector is often challenging because staff do not like them or are unfamiliar with their use. Additionally, Cabaobao

et al. [15] discussed how problems with system integration and the use of new technologies can hinder the successful application of automation. However, these obstacles can be overcome through careful planning, adequate training, and assurances from agency leadership to ensure the successful integration of digital systems.

The integration of digital transformation and automation in government agencies, as per the studies reviewed, plays a crucial role in enhancing efficiency, quality of service, and the government's responsiveness to the public. Malubag et al. [15], Syaidaturrahmi et al. [16], and Daud et al. [18] demonstrated that the use of modern technologies improves the operation, data accuracy, and accountability of government institutions, all of which are closely related to the development of a Centralized Client Information Management System (CCIMS). According to Jonathan [17] and Botangen et al.[3] state that digital systems such as web-based platforms and integrated data management tools promote accessibility, flexibility, and coordination among departments. According to Adewumi and Abasilim [23] and Cabaobao et al.[15], Some prevalent obstacles include employee resistance to change, a lack of digital skills, and issues with systems integration. Despite this, studies suggest that these challenges can be addressed with proper training, effective leadership, and cautious planning. As a result, the success of digital transformation is determined not only by a strong technology infrastructure but also by the efficiency and readiness of those involved in its implementation. For CCIMS, successful adoption is dependent not only on current technology but also on providing sufficient skills and assistance to government personnel to utilize the system's features effectively. Overall, studies indicate that when implemented effectively, digital systems such as CCIMS can enhance government operations, improve the quality of public services, and promote transparency a step toward more effective governance that benefits citizens and other sectors of society.

Digital Literacy and Data Management in Public Sector Service

The Report on Digital Literacy and Data Management in Public Sector Services highlights the importance of digital literacy and data management in enhancing efficiency and transparency in government operations and public services. It discusses the critical role of statistical literacy and digital systems in transforming public sector services, especially in monitoring and managing clients, decision-making, and improving overall services, such as the CCIMS system of the Butuan City Social Welfare and Development Department.

Several studies have emphasized the importance of digital literacy and data management in improving public sector services. For example, Koga [24] demonstrated that statistical literacy is crucial in reviewing statistical reports and emphasized its importance in public services, where the interpretation of data is critical for informed decision-making. Additionally, Simon [25] reviews the implementation of RA 9994 for seniors, emphasizing the need for stable data management systems to increase transparency and enhance services to government programs, in line with the goals of CCIMS-like systems. Meanwhile, in analyzing the Greek education system, Goulas [26]] demonstrated how data-driven insights help align the results of education with the needs of the labor market, highlighting the importance of effective data integration in public services. In addition, in his research on high school students in Japan, Koga [24]] re-emphasized the importance of statistical literacy skills, such as critical data analysis, for reading reports in various fields, including public welfare. In total, these studies demonstrate that statistical literacy and proper data management are crucial for improving efficiency, transparency, and accountability in public services, as shown by CCIMS.

Studies demonstrate how statistical literacy and data management systems enhance the efficiency of government services. For example, Koga [24] has shown that statistical literacy enables individuals and officials to understand and evaluate data, which is directly related to the management of CCIMS social welfare data. Simon [25] discovered that government programs, such as RA 9994, benefit from proper data management systems to enhance service, which aligns with the CCIMS's goal of making client data management more effective.

Although studies generally show that digital literacy and data management systems have positive effects, they also present some challenges. According to Koga [24], even with designated courses, students may still struggle to understand complex statistical data without continuous guidance and training. Similarly, Goulas [26] found that in some instances, a gap exists between educational and labor statistics and actual public policy outcomes, suggesting that CCIMS needs to address integration issues to be fully effective.

Research studies have shown the critical role of digital literacy and efficient data management in enhancing the efficiency, transparency, and accountability of public services, as demonstrated by the Centralized Client Information Management System (CCIMS). Koga [24] emphasized the importance of statistical literacy in accurately interpreting data for informed decision-making, an aspect that directly aligns with CCIMS' goal of developing data management for social programs. In the same perspective, Simon[25] explained that having a robust data management system, such as CCIMS, brings a higher level of transparency and better service to government programs such as RA 9994 for older people. In addition, Goulas [26] and Koga [24] both emphasized the importance of utilizing data-driven insights and statistical skills to align public services with the ever-changing needs of society. However, some challenges remain,

particularly in ensuring that officials possess sufficient digital literacy to effectively use and understand the data, as Koga and Goulas emphasized.

The Role of Management Information Systems (MIS) in Education and Public Administration

The implementation of Management Information Systems (MIS) in governmental organizations, particularly in education and social welfare, is crucial for enhancing administrative efficiency, effective data management, and informed decision-making processes. These technologies facilitate the collection, storage, retrieval, and analysis of data, enabling institutions an increase their level of transparency, accountability, and responsiveness. This theme determines how MIS enhances the operational efficiency of educational institutions and social well-being systems through automation of processes, real-time data access, and improved decision-making.

Some studies have shown that Management Information Systems (MIS) play a significant role in improving the efficiency and quality of service in various industries. Zhang [27] demonstrated in his study of Learning Management Systems (LMS) how they enhance students' learning and the effectiveness of teaching by providing automatic supervision, progress monitoring, and personal resources. This concept is comparable to the CCIMS approach of Centralizing Clients' data on social services, which automates reports and organizes the allocation of resources to enhance program implementation. Similarly, Orhani et al. [28] evaluate how electronic management systems enhance communication, streamline workflows, and facilitate better decision-making the primary roles of CCIMS in the City Social Welfare and Development Department.

Meanwhile, Dahyanti and Hamini [29] emphasized the importance of MIS in school management through automatic data flow and facilitating decision-making, similar to CCIMS's role in reducing manual employment and increasing data accessibility for more effective social services. Triwiyanto et al. [30] note that MIS is increasing its responsibility to schools, which is directly related to the role of CCIMS in ensuring transparency in client data management and the allocation of social program resources. In addition, Godoy et al emphasized. Eventually, Zhang [27] demonstrated how MIS could be adapted using Deep Learning Algorithms to enhance information literacy. This method aligns with CCIMS's purpose of providing specialized data to staff for more informed decision-making based on each client's needs.

Created studies show that MIS systems, specifically those implemented by CCIMS, improve data management in governmental services. For example, MIS in schools automates administrative activities and enhances communication, as noted by Orhani et al. [28], which is comparable to the CCIMS's ability to focus on client data and automate reports. Studies also emphasize the importance of user-centered design to make CCIMS-like systems accessible and effective for end users.

Although the benefits of MIS are clear, there are also challenges in studies. For example, Zhang [27] noted that if there were no appropriate learning routes, MIS could fail in education to meet the specific needs of students. In the same context, Triwiyanto et al. [30] found that inadequate training and infrastructure in CCIMS can prevent social well-being professionals from fully utilizing the system. The success of MIS implementation depends not only on technology, but also on the staff's preparation and the institution's support.

All in all, Management Information Systems (MIS), such as CCIMS, are innovative tools that can significantly improve efficiency, transparency, and accountability in public sector operations. These

systems enhance workflow, offer real-time access to data, and facilitate data-driven decision-making. However, for the MIS to be genuinely effective, adequate training, infrastructure, and institutional support are necessary. Studies have proven that when the MIS is implemented, considering the needs of users and the organization's readiness, it can significantly improve service delivery, as demonstrated in its applications in education and social well-being.

The use of centralized database systems in both public and private sectors has proven to enhance efficiency, accuracy, and decision-making skills. These systems integrate various types of data on a single platform, allowing for real-time updates and easy access. In public services such as social and education well-being, centralized databases ensure that essential managers and service providers have quick access to the correct information. This topic examines the use of centralized database systems in the aviation industry and in the management of social well-being, emphasizing how they contribute to operational efficiency, data integrity, and the improvement of service delivery.

Tarmidi et al researched. [31] The development of a web-based orphanage database system for the coat in Harang, Malaysia, aimed at encouraging information about orphanages, residents, and external stakeholders. This solution solved issues such as slow information extraction, data repetition, and security threats caused by paper-based records. By transferring to the digital system, the ornament has improved data accessibility, transparency, and decision making - similar to the CCIMS method in clocking client data to improve operations efficiency in the City Social Welfare and Development Department. In the same way, Sridhar et al. [32] utilized a centralized database system to manage plane reservations and ticketing. Their studies have found that centralized databases increase accuracy in flight availability, eliminate booking errors, and provide real-time updates critical aspects for large-scale

operations. Like the CCIMS, which unites various services in social well-being through the centralized database, the Airline system also uses APIs to facilitate continuous interaction between airlines, agents, and passengers, resulting in increased transparency and operational efficiency.

Centralized Database Systems in Public and Private Sectors

The use of centralized database systems in both public and private sectors has proven to enhance efficiency, accuracy, and decision-making skills. These systems integrate various types of data on a single platform, allowing for real-time updates and easy access. In public services such as social and education well-being, centralized databases ensure that essential managers and service providers have quick access to the correct information. This topic examines the use of centralized database systems in the aviation industry and in the management of social well-being, emphasizing how they contribute to operational efficiency, data integrity, and the improvement of service delivery.

Tarmidi et al. did some studies [31] The goal of creating a web-based orphanage database system for the coast in Harang, Malaysia, was to make it easier for people to share information about the orphanage, its residents, and outside partners. This answer fixed problems like getting information slowly, having duplicate data, and security risks that come with keeping records on paper. By switching to a digital system, the ornament has made data easier to reach, more clear, and better decisions. This is similar to how the City Social Welfare and Development Department uses the CCIMS method to keep track of client data to make operations run more smoothly. In the same way, Sridhar et al. [32] used a central computer

system to handle booking and buying plane tickets. Their studies have found that centralized databases increase accuracy in flight availability, eliminate booking errors, and provide real-time updates —critical aspects for large-scale operations. Like the CCIMS, which unites various services in social well-being through the centralized database, the Airline system also uses APIs to facilitate continuous interaction between airlines, agents, and passengers, resulting in increased transparency and operational efficiency.

Both studies emphasize the essential benefits of using the centralized database system in their respective industries. Tarmidi et al. [31] developed an orphanage database system that centralizes clients and operations, providing rapid access, increased data security, and improved resource allocation for operational purposes. Meanwhile, the airline reservation system used by Sridhar et al. [32]. A centralized database to synchronize the schedule of flights, bookings, and ticketing in real time, ensuring accuracy and efficiency at all stages of service. Both studies emphasize the importance of centralization of data for better decision-making, operational efficiency, and transparency.

While both systems utilize a centralized database to increase efficiency, their scope and purpose differ. The Orphanage Database System is primarily designed to manage clients and resources for social programs dedicated to serving people in need. On the other hand, the airline reservation system stores transactional data at a higher level, which necessitates real-time flight availability and booking information exchange among various stakeholders, including airlines, travel agents, and passengers. These two examples demonstrate the flexibility of centralized database systems, which can be tailored to meet the specific needs of various industries.

Centralized database systems are innovative tools in both social and commercial well-being sectors, as they provide epic data management, real-time updates, and improved service delivery. CCIMS can learn

from the real-time data processing of the airline reservation system, as well as the emphasis on accessibility and transparency in the orphanage database system. Both studies emphasize the importance of centralized databases in enhancing operational efficiency, improving data security, and facilitating better decision-making across various industries. However, the success of these systems depends not only on the technical infrastructure but also on user acceptance, system integration, and ongoing support.

1.3 Objective of the Study

The researchers' study aims to design, develop, and implement a Centralized Client Information Management System for the Butuan City Social Welfare Development Department, named Centralized Client Information Management System (CCIMS).

Specifically the study aims to:

1. Design a role-based login system that provides super administrator, division administrators and encoders separately, safely, and in accordance with their duty accounts.
2. Integrate data visualization and reporting tools that allow CSWDD staff to generate comprehensive reports on the total number of beneficiaries served, categorized by program, barangay, year, and month, to support data-driven decision-making.
3. Implement a notification feature that allows the administrator to receive timely alerts regarding new client registration, data update, and other important system activities.
4. Evaluate the system’s functionality, efficiency, and usability using ISO 25010 standards.

1.4 Scope and Limitations

The “Centralized Client Information Management System (CCIMS)” project focuses on enhancing and streamlining the reports and processes at City Social Welfare and Development Department (CSWDD). In

this system, the administrator is responsible for account and client management, while the encoder is responsible for inputting data information of the clients. CCIMS will be a web-based platform accessible through desktop computers and laptops connected to the internet.

However, the system has limitations. It will not include features like financial tracking, budgeting tools, biometric login, or mobile app access. The project is also limited by time and available resources, so advanced features like predictive analytics or AI-based decision support will not be included in this version. The system requires a stable internet connection to function properly, and staff members unfamiliar with digital tools may need training. Data migration from existing spreadsheets will be performed manually, which may take time and require additional verification to ensure accuracy.

1.5. Definition of Terms

Client Information - refers to the personal, demographic, and service-related data encoded into the CCIMS platform. It includes names, addresses, program enrollment, and service history, which social workers use to track client progress, generate reports, and ensure accurate delivery of welfare services.

CAR – Children At Risk program.

CICL – Children In Conflict with the Law program.

CIU – Crisis Intervention Unit

COD – Community Operations Division

CSAE – Child Sexual Abuse and Exploitation.

Data Breach - refers to any unauthorized access or exposure of client records stored in the system. The project includes encryption, role-based access, and audit trails to prevent breaches and protect sensitive information from misuse or accidental disclosure.

Database Management System - to store, organize, and retrieve client data efficiently. It ensures data integrity, supports multi-user access, and enables automated queries for generating reports and monitoring program performance across different offices.

DPSD – Development and Promotive Services Division.

End-user - refers to the person that will utilize and interact with the CCIMS, such as the CSWD Social Workers - the super admin, division admin, and encoder as an ordinary users' who can only interact with the system by monitoring all client information.

HDD - use on legacy systems or backup servers where CCIMS data is stored. While slower than SSDs, HDDs provide large-capacity storage for archival purposes and system logs, especially in offices with limited hardware resources.

ISO/IEC 25010 - An international software quality model was used in this study to guide testing of the System's functionality, reliability, performance efficiency, usability, security, maintainability, and portability. In this study, ISO/IEC 25010 serves as the benchmark for testing and evaluating the CCIMS. It ensures the System is not only functional but also efficient, secure, user-friendly, and sustainable in the long run.

OSCA – A division under CSWDD called Office of the Senior Citizens Affair.

PDAO – A division under CSWDD called Persons with Disability Affairs Office.

PSSD – PWD AND SENIOR CITIZEN SERVICES DIVISION.

PWUD – Person Who Used Drugs.

Role-based- refers to a system access control approach where users are assigned specific roles (e.g., super admin, division admin, office staff, services encoder), and each role determines the level of access and privileges a user has within the system.

RPSD – Restorative and Protective Services Division

Social Welfare - refers to the services and programs managed by the Butuan City DSWD to support vulnerable populations. The CCIMS is designed to enhance the delivery of these services by improving data accuracy, tracking client needs, and enabling evidence-based decision-making.

SSDs - are preferred for hosting the live database and web application due to their speed and reliability. They enable faster data retrieval, reduced system downtime, and improved performance for users accessing client records and generating reports.

Statistical Report - a document generated by the system that represents numerical data in a structured format. In this project, statistical reports include summaries, charts, or tables that help the organization analyze trends, measure performance, and support decision-making.

User Interface (UI) - the visual and interactive part of the system allows users to communicate and interact with the application. It includes menus, buttons, forms, and other design elements that make the system easy to navigate and operate.

VAWC – Violence Against Women and Children.

CHAPTER 2

METHODOLOGY

2.1 Research Design

This study will use a Developmental Research Design (Systems Development Research)**.** It focuses on designing, developing, and evaluating the Centralized Client Information Management System for Butuan City Social Welfare Development Department. The research will follow the Design and Development (D&D) method**,** ensuring the System's functionality and usability.

2.2 Software Development Life Cycle



Figure 1. Agile Methodology

As depicted in Figure 1, our system will follow the agile methodology to ensure incremental, flexible, and user-centered development. Through this iterative process the system will be designed, ideally developed, reviewed, and refined based on actual feedback from CSWDD staff. Through this approach, improvements can be made faster, errors are detected early, and the final system becomes more efficient, user-friendly, and aligned with the department's actual operations.

2.3 System Architecture

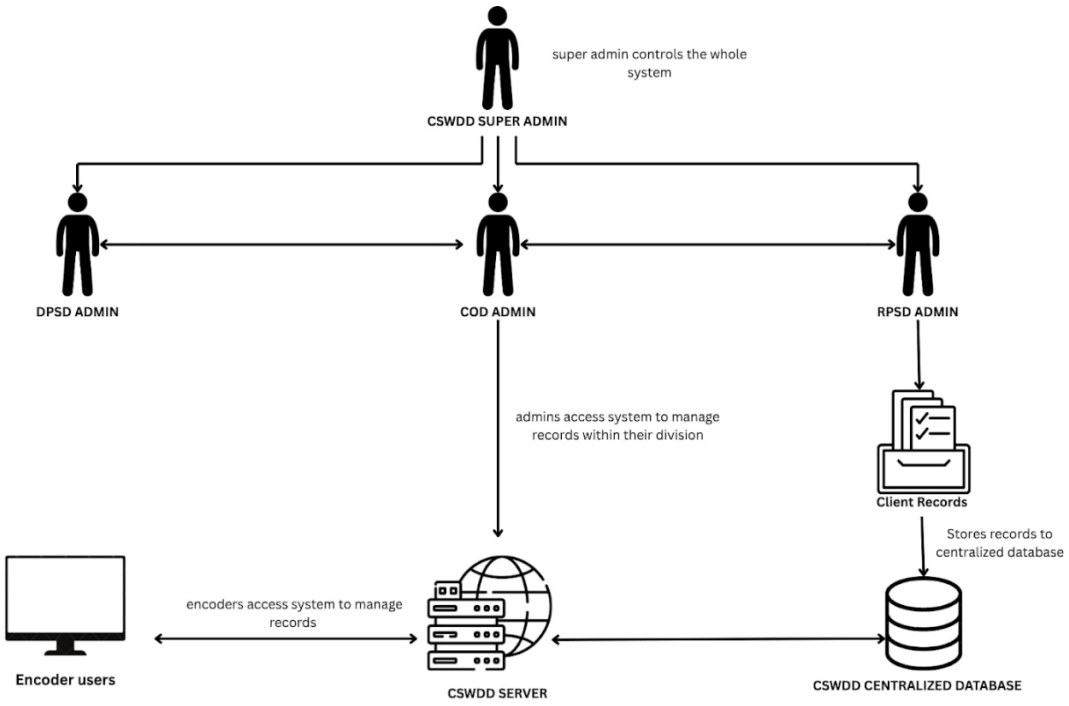


Figure 2. System Architecture

Figure 2 shows how the CSWDD Centralized Client Information Management System (CCIMS) is built and how administrative tasks and encoder users deal with centralized databases. The CSWDD Super Admin is at the top of the hierarchy. They are in charge of the whole system and make sure that users can get into sections. The DPSD admin, the PSSD admin, and the RPSD admin are shown below. They are all in charge of keeping client data in their own divisions. Encoder users are accessing the

system via the CSWDD server to input, update, and manage client data, which is then stored in CSWDD's centralized database. This structure makes sure that data is handled safely while keeping management centralized. This promotes consistency, efficiency, and easy reporting to all units.

2.4 Conceptual Diagram (IPO Diagram)

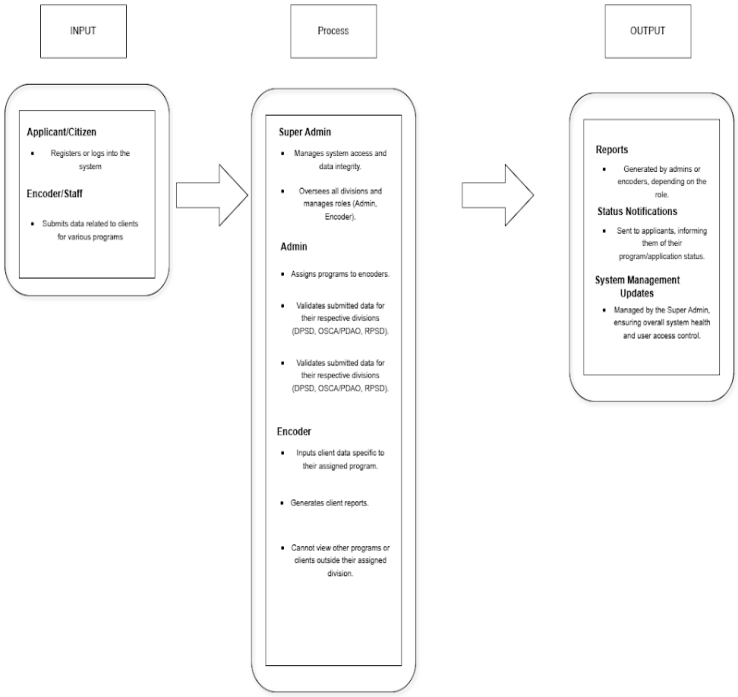


Figure 3. Conceptual Framework (IPO Diagram)

Figure 3 illustrates the system's flow in the CCIMS-BCSWDD conceptual diagram, which begins with applicants (citizens) registering or logging in to apply for various programs, including RPSD, OSCA, or those for senior citizens. After logging in, the encoders create reports and add comprehensive information on the clients they have been assigned; however, they are only able to manage the clients in their particular program. The administrators oversee departments, verifying the information encoders submit and producing reports that show the development of their own initiatives. Admins are also in charge of giving encoders and making sure that all the data is correct and complete. The Super Admin has full access to the system and decides what jobs and permissions users have. They also keep an eye on the whole process and make sure the system is safe and working. Reports that show the state of clients and programs, messages telling applicants how their applications are going, and system management updates from the super admin to make sure the system works right are all things that the system sends out. This flow not only keeps candidates informed but also makes sure that all the data in the CCIMS platform is correct. It also makes sure that each job has clear duties and that the system works well.

2.5 Use Case Diagram

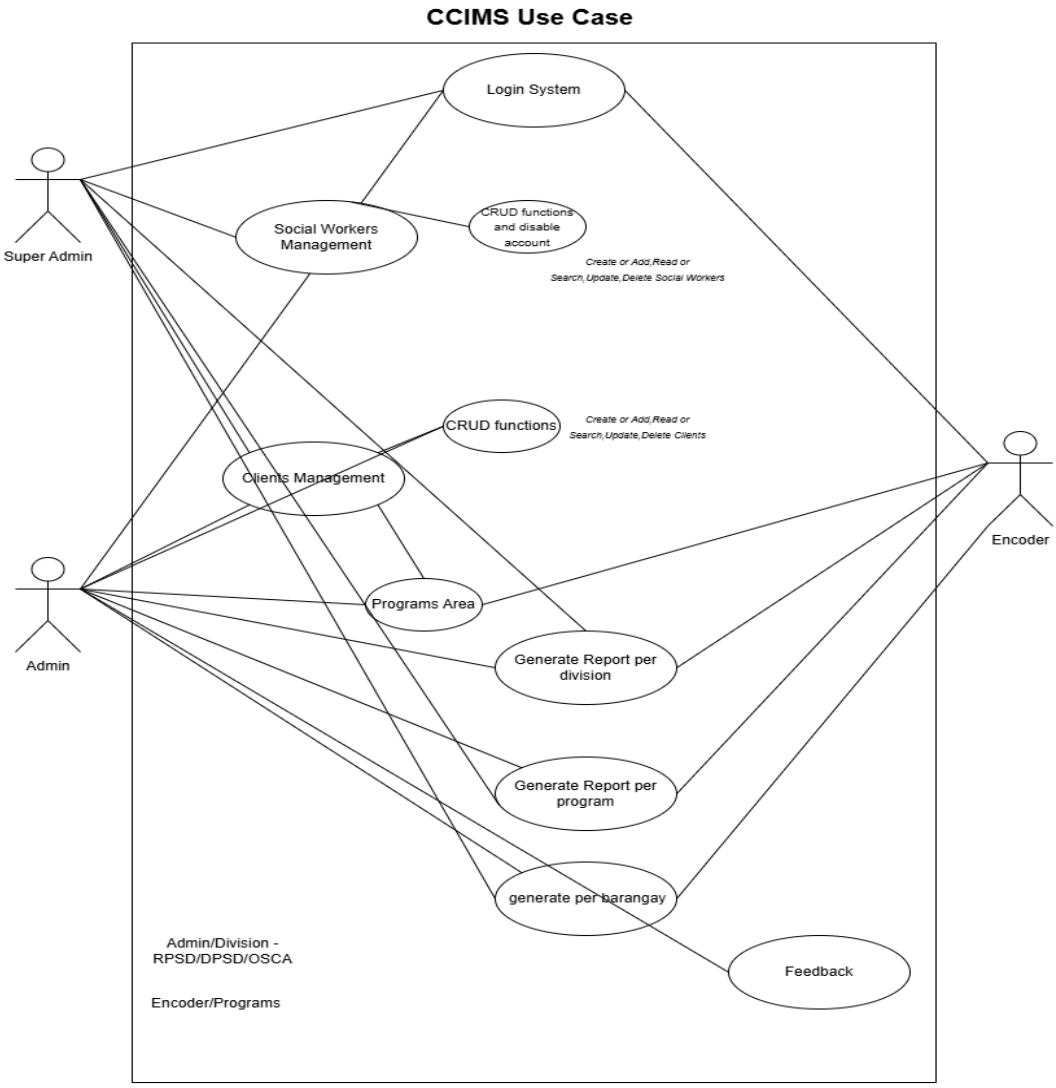


Figure 4. Use Case Diagram

Figure 4 shows the relationship of actors and activities (use cases) within the system. The Super Admin has maximum control over the system, with the ability to log in, manage social workers, and perform CRUD (Create, Read, Update, Delete) functions, as well as deactivate accounts. He also has access to the management of clients and encoders, as well as programs under each division. Super Admin has extensive control over essential system components.

The admin has similar responsibilities to the Super Admin, but he focuses on each division's programs, such as RPSD, DPSD, and PSSD. The admin can also generate reports for each division, program, and barangay. The admin's tasks are focused on auditing and reporting the progress of the programs managed by his division.

The encoder, in turn, has the primary responsibility for managing programs, specifically the Encoder Programs. It can perform CRUD (Create, Read, Update, Delete) functions to update and manage client data. The encoder's responsibility also includes generating reports related to programs and client information under their division.

In the Reports section, the system provides the ability to generate reports that help monitor the progress of programs managed by the Admin and Encoder. These reports are focused on each division, program, or barangay and are used to ensure transparency and accountability in their activities.

The last part of the diagram is the feedback, where the encoder is connected to the feedback system. The feedback loop is essential for understanding the opinions or input of clients or other users regarding the services and programs they have encountered. The overall diagram illustrates the orderly flow of activities and responsibilities of each role in CCIMS, from managing clients to analyzing reports and feedback from system users.

2.6 Activity Diagram

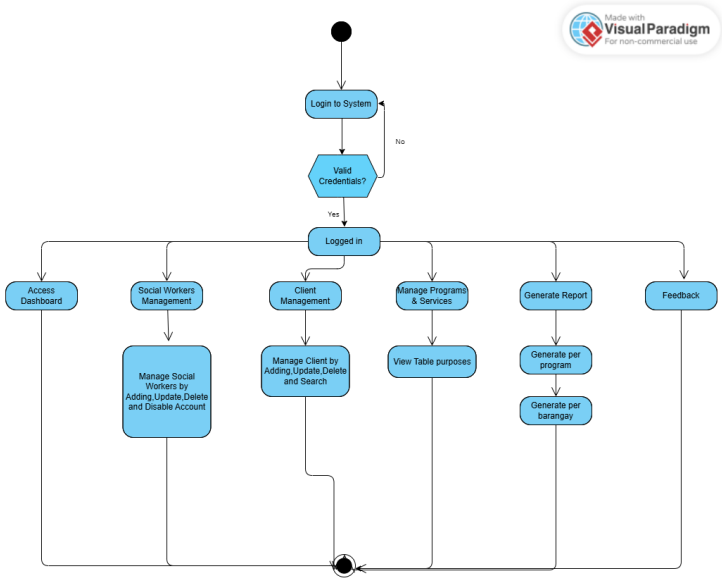


Figure 5. Activity Diagram (Super Admin)

Figure 5 shows the most important things that the Super Admin does in the CCIMS system. The first step is to log in to the system. This is where the user types their credentials to be checked. After the passwords are checked to make it sure they are correct, the Super Admin can go to the system dashboard and do things like manage users, which lets them make, change, or delete user accounts.

Following this, the Super Admin will proceed to Client Management, where clients' information is managed. It includes viewing and updating client information, as well as assigning them to relevant programs and services. Then, the Super Admin will continue to manage programs, where he can create, edit, or delete programs in the system.

In the next step, the Super Admin can also generate or view reports related to the system, as well as Audit Logs to see the activities within the system. The last step performed by the Super Admin is to Track System Health, where the health and state of the system are monitored to ensure it is running correctly.

If the credentials passed during login are incorrect, the system will display an error message, and the process will end. This diagram illustrates a precise flow of tasks and steps that the Super Admin must follow to ensure CCIMS runs smoothly.

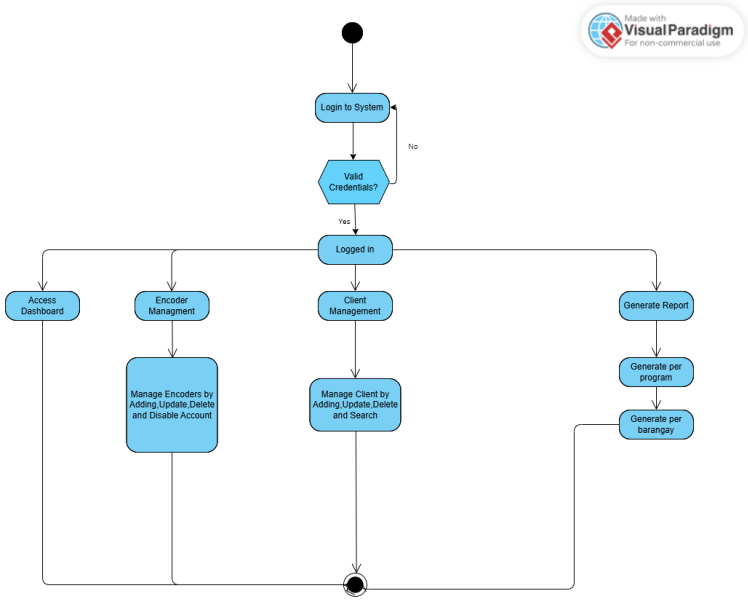


Figure 6 Activity Diagram (Admin)

Figure 6 shows the steps followed by Admin in CCIMS system. The process starts with the Admin login to the system, where the credentials will be entered to be checked by the system. Once the credentials are correct, the Admin can enter the dashboard and start tasks such as managing encoders and programs. The Admin has the ability to manage the programs and encoders that are part of their responsibilities.

After that, the Admin will continue to manage the clients, where he can view and update the information of the clients. This includes assigning clients to programs that focus on their needs. Admin can also generate reports related to programs and clients. He can generate reports for each division, program, and barangay to monitor the progress and status of these programs.

These steps also include Audit Logs, where the Admin can view the activity logs to track all actions performed on the system. The Admin also has the ability to monitor the health of the system and ensure that it is running properly through Track System Health.

If the credentials entered are incorrect, the system will display an error message and the process will end. This diagram illustrates the proper flow of tasks that the Admin needs to perform to ensure the proper operation of the CCIMS system.

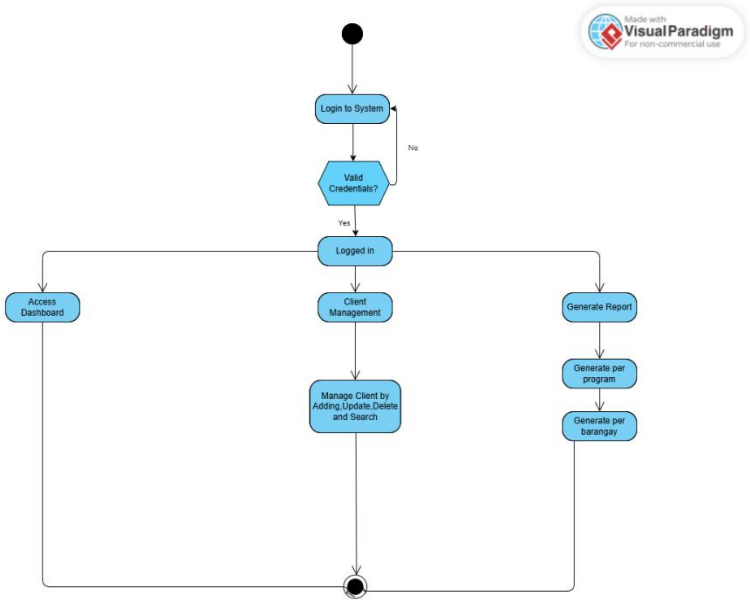


Figure 7 Activity Diagram (Encoder)

Figure 7 shows the steps that the encoder follows in the CCIMS system. The process begins with logging in to the system, where the user enters their credentials for verification by the system. Once the credentials are correct, the encoder will be able to enter the dashboard and start managing clients' information. The encoder enables users to view and update client details within the system. It includes adding clients who have been serviced or received services from the program.

The encoder can also make reports about barangays and projects. To keep track of success and make sure services are being given properly, they can make reports based on program details and clients in each barangay. The system will show an error message and end the process if the passwords are wrong. This picture shows the important steps and jobs that the encoder needs to do to make sure the CCIMS system works right.

2.7 Sequence Diagram

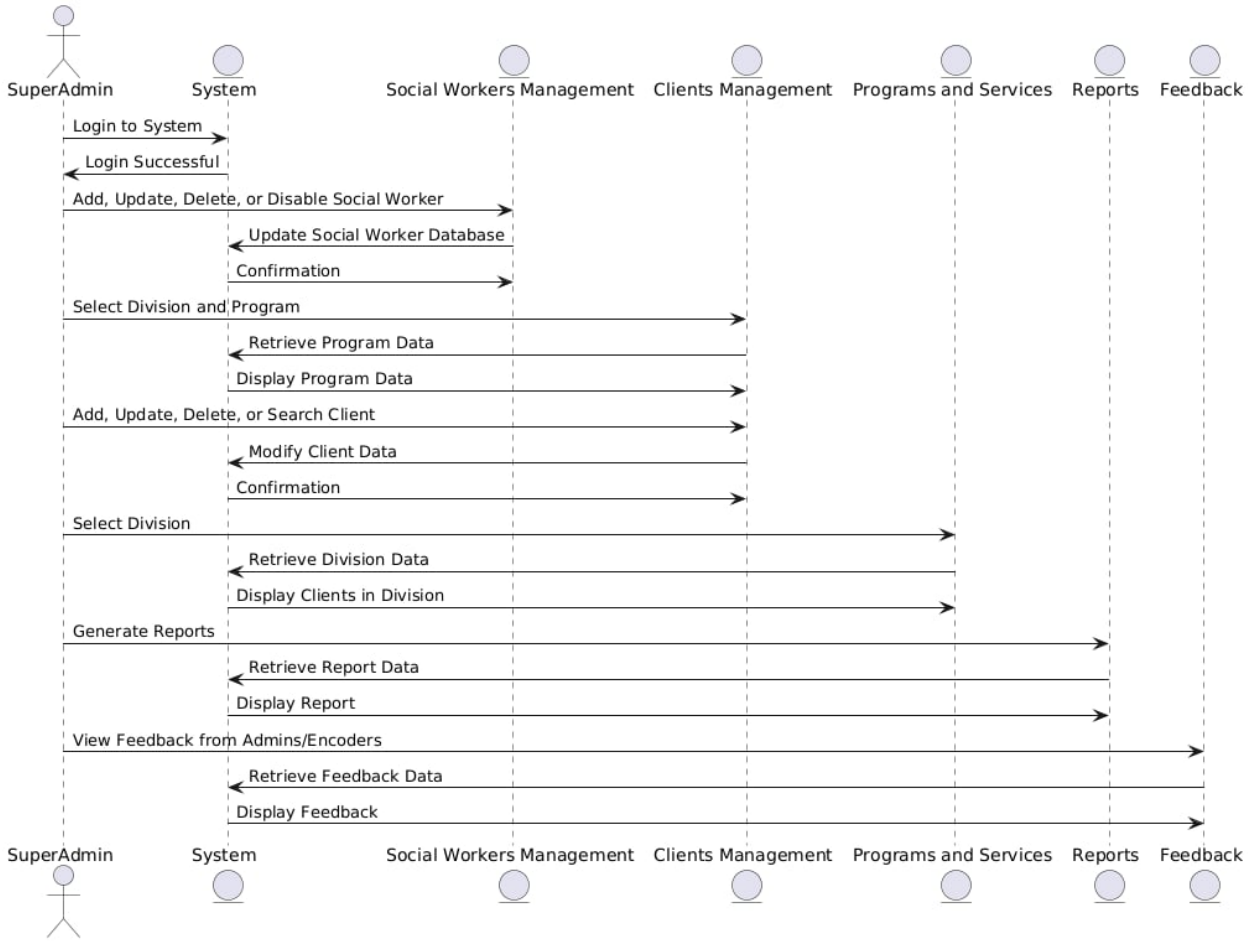


Figure 8. Sequence Diagram (Super Admin)

Figure 8 shows the Super Admin workflow while using the system. It starts with logging in to the system using the proper credentials. If the login details are incorrect, the system will ask to replace them with the correct information. Once logged in, the Super Admin can manage social worker data, such as adding, updating, or deleting information. After each action, the system confirms changes in the social worker database.

After that, the Super Admin will select the division and program, and the system will automatically retrieve the relevant program data and display it. Next, the Super Admin can manage client details by adding, updating, removing, or searching for a specific client. Once the changes are completed, the system confirms that client data has been updated.

The Super Admin may also select the division, which will instruct the system to retrieve and display data specifically for that division. Another Super Admin task is to generate reports, in which the system captures the required data and displays it.

The Super Admin can see what admins and encoders have written in the last part, and the system will show any feedback that has been sent in. This process makes sure that the Super Admin can easily find their way around the system and keep an eye on everything from start to finish.

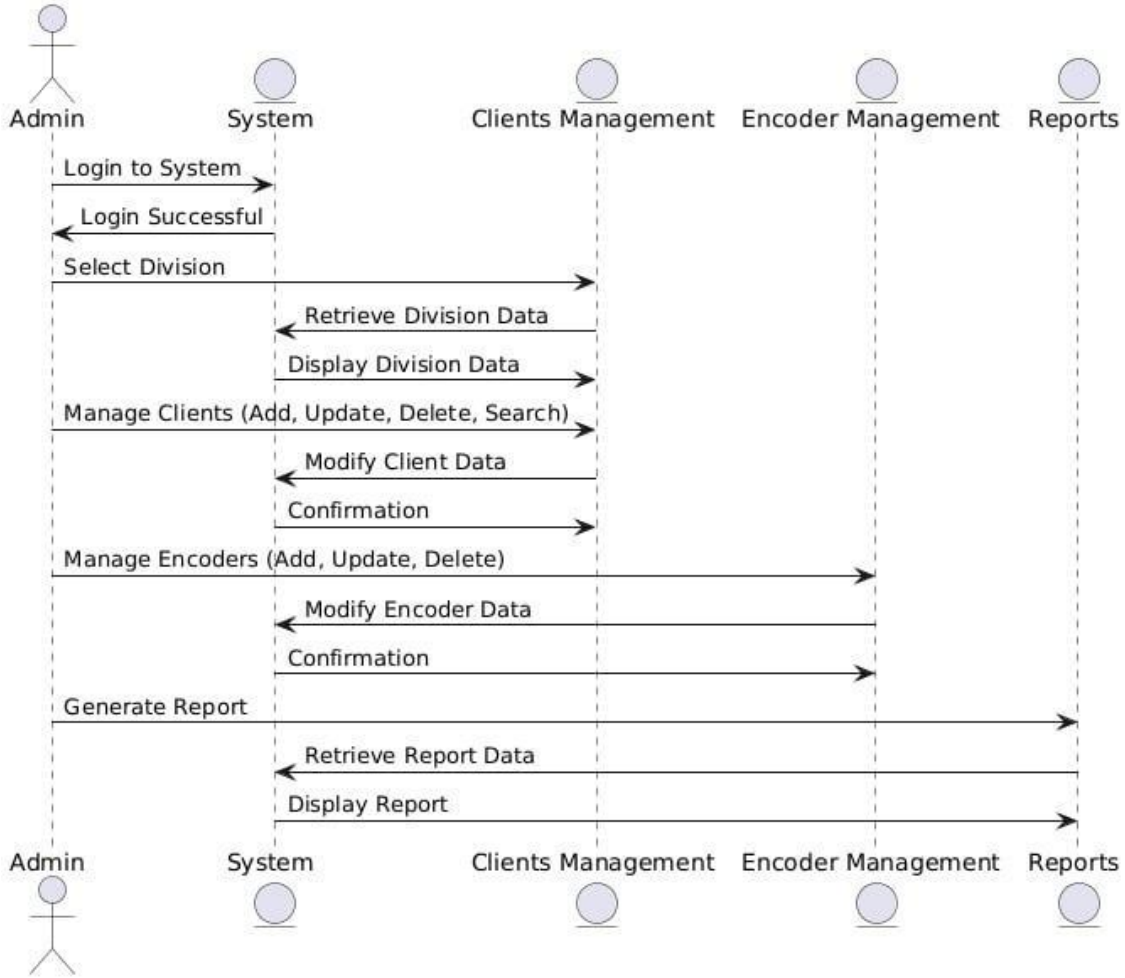


Figure 9. Sequence Diagram (Admin)

Figure 9 shows the UML's adoption diagram for the CSWDD Centralized Client Information Management System (CCIMS), detailing a relationship between the admin and various system modules. Following this, the next step starts with the admin's system swing, followed by a successful verification response. When lying in, the admin selects a division, which motivates the system to obtain and display the data. The admin then performs client management tasks, such as adding, updating, removing, or finding records, which are processed by the client management module. Similarly, encoder data is managed through the encoder management module, which supports administrative control over user access and duties. Finally, the admin develops reports, which enable the system to obtain and display data

from the report module. This diagram features a series of flow operations and confirms how each module responds to administrative actions, ensuring a structured and responsive user experience.

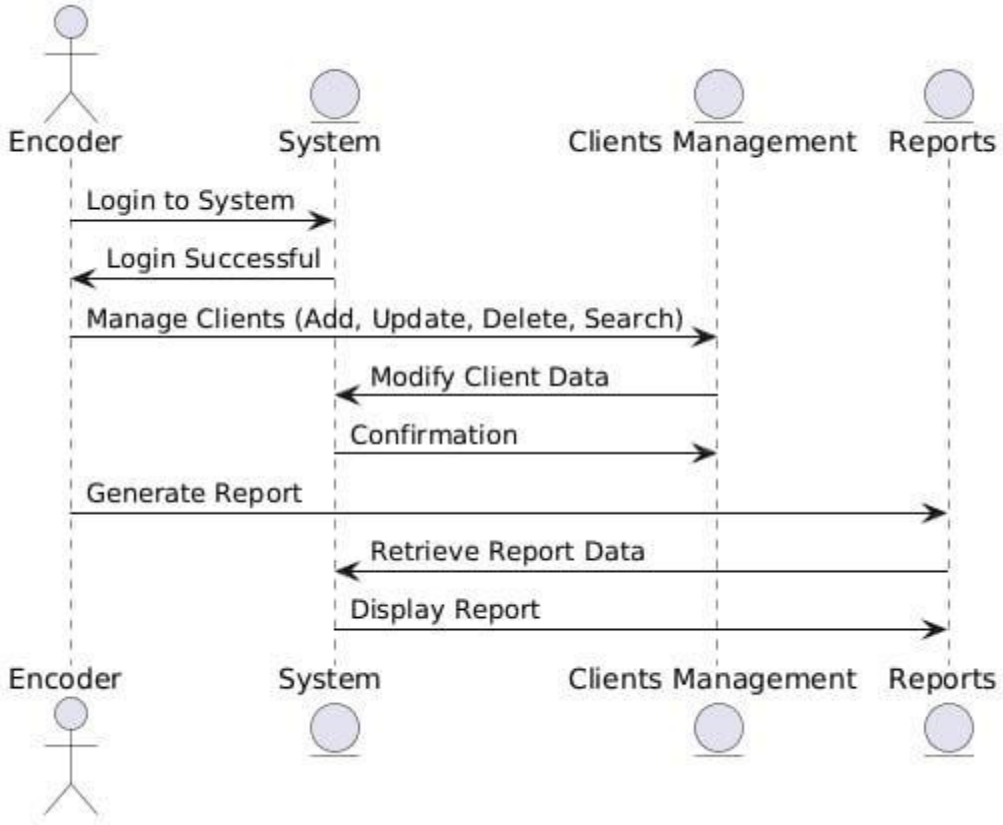


Figure 10. Sequence Diagram (Encoder)

Figure 10 shows the UML order diagram for Encoder contacts in the CSWDD Centralized Client Information Management System (CCIMS). It shows the order of processes for managing clients and making reports step by step. The next step happens when the encoder swings into the system, and it ends with a successful verification response. When the encoder is logged in, it does tasks connected to the client, like adding, updating, removing, or finding records. These tasks are handled by the client management module and approved as needed. After that, the encoder sent a request to make the Report. This made the system get data from the report module and show the results. This picture

shows how the encoder works on paper and how the modular system response structure improves use, response, and system support for correct data handling and reporting.

2.8 ERD (Database Design)

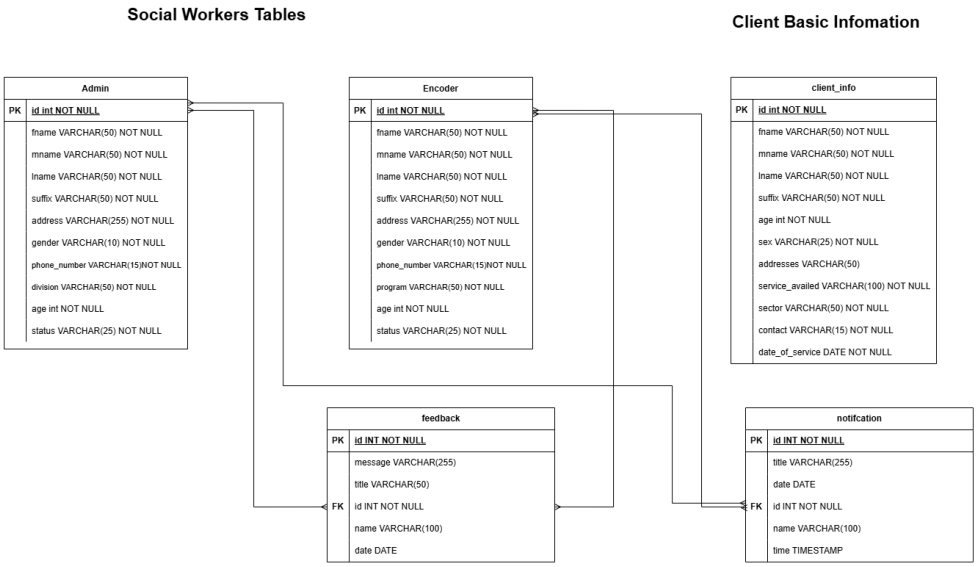


Figure 11. ERD Database Design

Figure 11 shows the structure of the CSWDD Centralized Client Information Management System (CCIMS) database, which focuses on the connection between data about the social worker and data about the customer. There are three key parts to the schema: admin, encoder, and basic client information. Personal information like name, address, gender, and phone number are stored in both the admin and encoder tables. Each table has its own special fields, such as "division" for the admin and "program" for the encoder, which show what services they are responsible for. The basic information table for clients is

where most of the client notes are kept. This is where you can find out about the client's demographics, service types, industry classification, and date of service. Supporting tables for notifications and comments make it easier for social workers and client workers to talk to each other and collect data. Foreign key relationships connect these records to their users. The picture mostly shows a well-organized relational database that keeps data safe, makes keeping records easy, and allows for a thorough review of user roles and client services.

3.6. User Interface Design (Prototype)

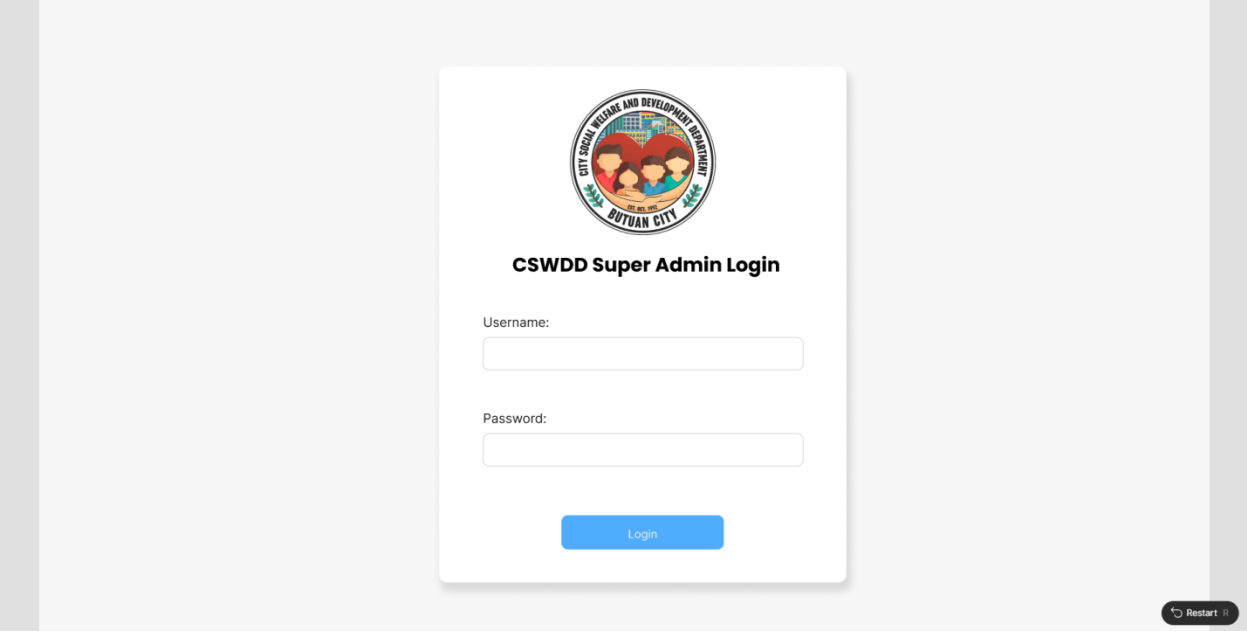


Figure 12. Login Page (Super Admin)

Figure 12 shows the CSWDD Super Admin interface, which is made to give the City Social Welfare and Development Department safe access to the management backend of a system they are likely to use. A circle mark at the top of the page builds institutional identity and trust by showing that the system is officially connected with the government. The design is simple and clean, drawing attention to the two main areas where users enter their information: username and password. When the login button is clicked, these fields collect credentials from approved users. The credentials are then checked against a

secure database. If the details match and the user that has Super Admin rights, they can use advanced system functions like managing data, running reports, and managing other users. Design usually comes before clarity, security, and ease of use. This makes sure that only super managers can be checked out and get to private parts of the CSWDD system.

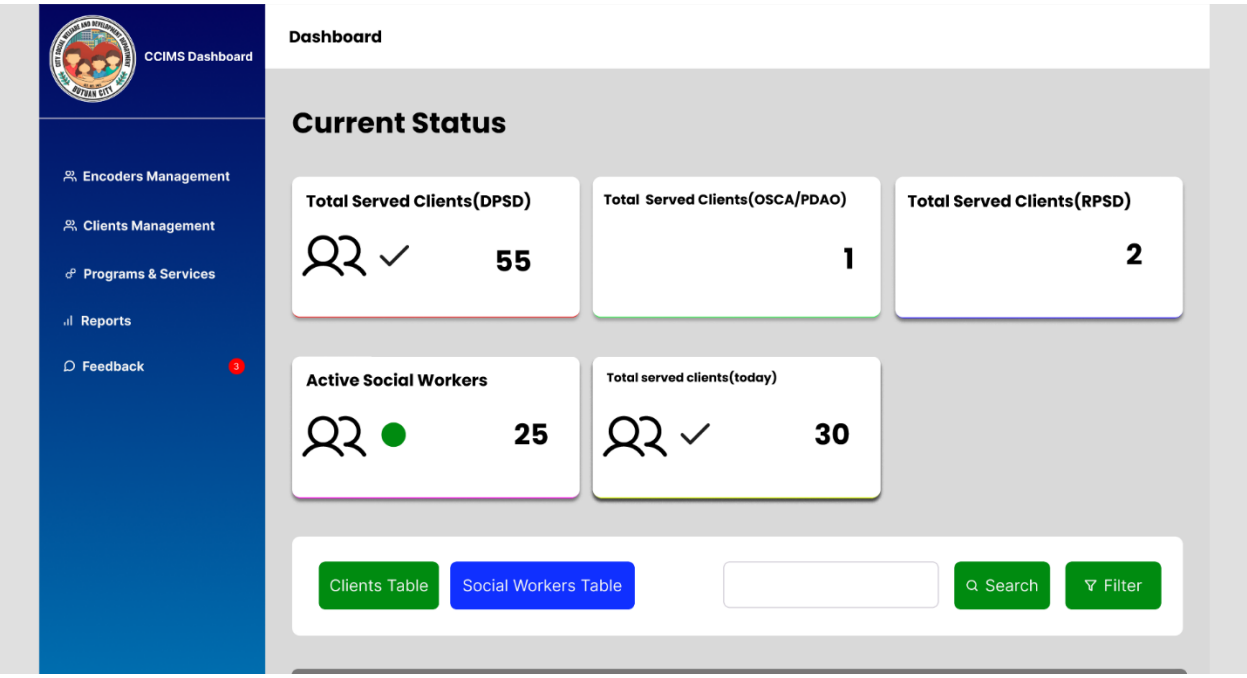


Figure 13. Super Admin Dashboard

In Figure 13 The CCIMS Dashboard is a centralized interface designed to provide real-time views on client-worker's client services and activities within the city's Social Welfare and Development Department. Its layout is structured for clarity and operating efficiency, with major metrics shown at the top - as the total number of clients served in various divisions (DPSD, PSSD, and RPSD), the number of active social workers, and the number of clients served in the present day. These figures offer administrators a quick snapshot of service delivery performance and workforce interaction. On the left side, a vertical navigation menu arranges core system modules including managing encoders, client management, programs.

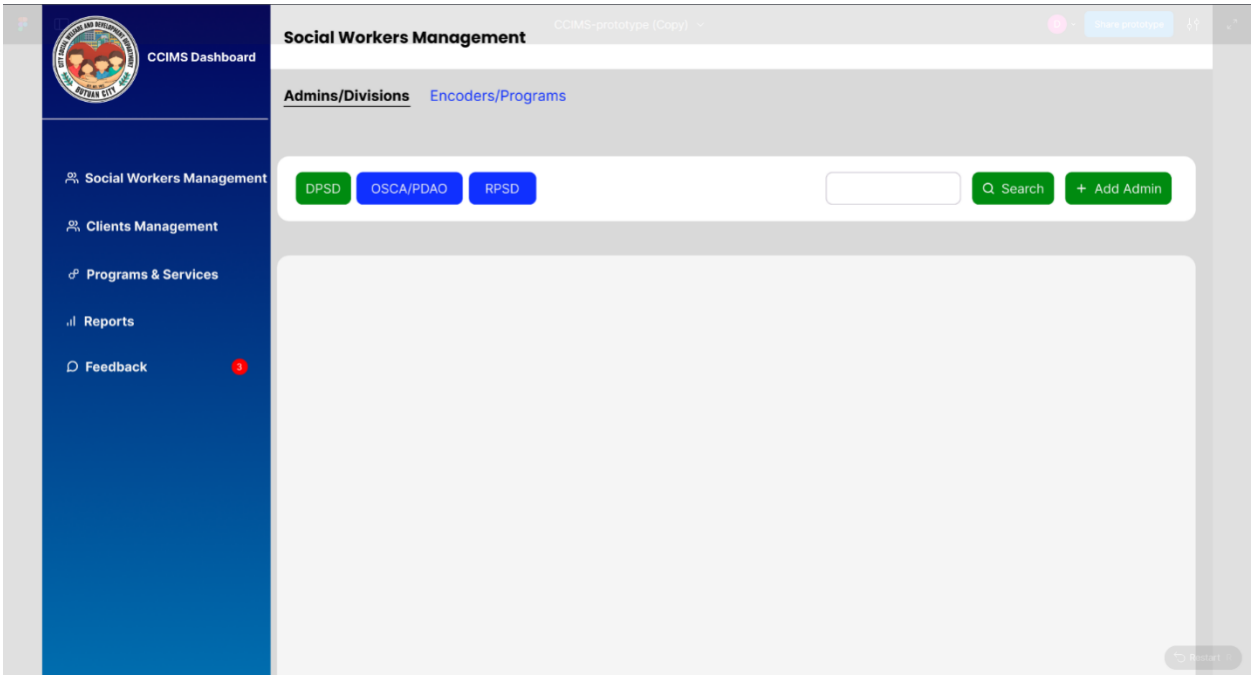


Figure 14. Super Admin account management

In Figure 14 The CCIMS dashboard for the management of society workers is designed to – streamline administrative administration and coordination with divisions within the city's Social Welfare and Development Department. The interface features a left navigation panel that arranges major modules such as social workers management, client management, programs.

Three known buttons labeled DPSD, PSSD, and RPSD allowed users to filter or switch views between different divisions or program areas, supporting target administration. On the right, a search bar and a green search button provides users to find specific records well, while the Add Admin's ADD button facilitates the onboarding of new administrative users. The top of the interface includes a circular logo that reinforces the institutional branding, and a user profile icon labeled "development," suggesting that the current user has access to the developmental level or part of the technical system group. Generally, the dashboard design emphasizes clarity, modularity, and control - empowering administrators to manage personnel and programs with precision and ease.

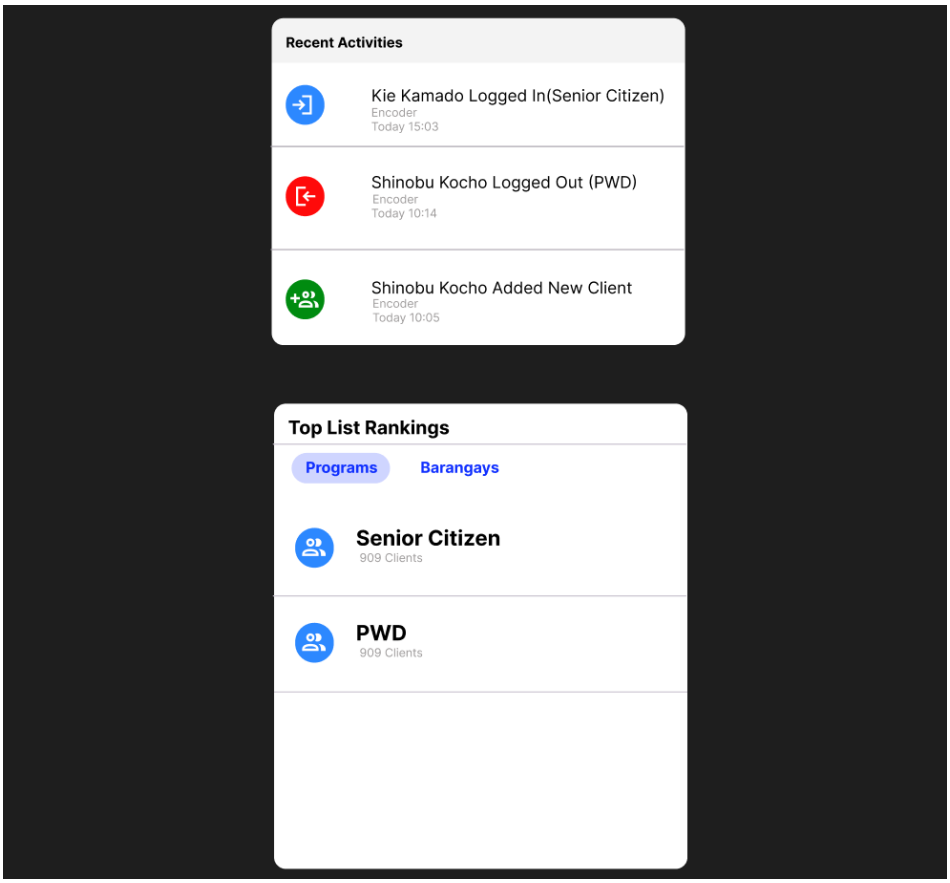


Figure 15. Performance Tracking

In Figure 15 This interface presents a dual-function layout that improves both responsibility and performance monitoring within the CCIMS system. On the left side, recent sections of activities log real-time actions performed by designated personnel, such as logins, logouts, and client registrations. Each entry includes the name of the social worker, their associated program (e.g., Senior Citizen, PWD), and a timestamp, allowing administrators to monitor who is actively interacting with the system and when. This feature supports transparency and helps monitor the user's relationships for auditing or support purposes.

At the bottom, the top rankings list section provides a visual summary of the program performance. In tabs for programs and barangays, users can be detained between views to see which categories or locals serve most clients. In the current perspective, both senior citizens and PWD programs

are tied to 909 clients each, indicating high contact or demand. This ranking system helps administrators identify which services are most used and may be advised to provide resources, staff, or outreach techniques. Together, these sections create a dynamic dashboard that mixes the user's responsibility with program level views.

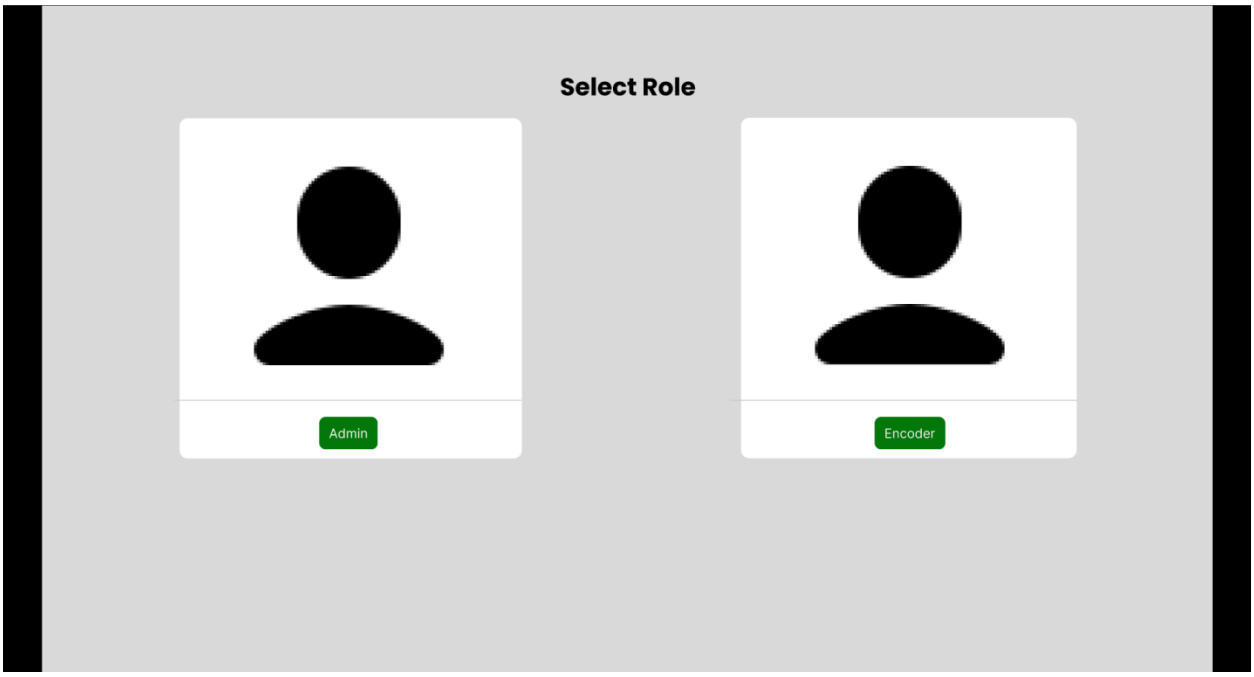


Figure 16. Select Role

In this Figure 16 The Select Role Interface serves as a gateway for users to specify their access level within the CCIMS system. Position at the top center, the title "Select Role" clearly communicates with the purpose of the screen. Below this, the two unique cards represent the available user duties: admin and encoder. Each card features a common user icon and a green button labeled with its own paper, allowing users to select their operating using a single click. The admin role is likely to provide access to high-level system controls, such as user management, program management, and generation reporting, while Encoder paper is customized for data entry tasks, such as registering clients and updating service notes. The clean, symmetrical layout and light gray background create a neutral, professional

environment that emphasizes clarity and ease of use. This design ensures that users can quickly identify and choose their role, streaming the login process and strengthening the control-based control of the role throughout the system.

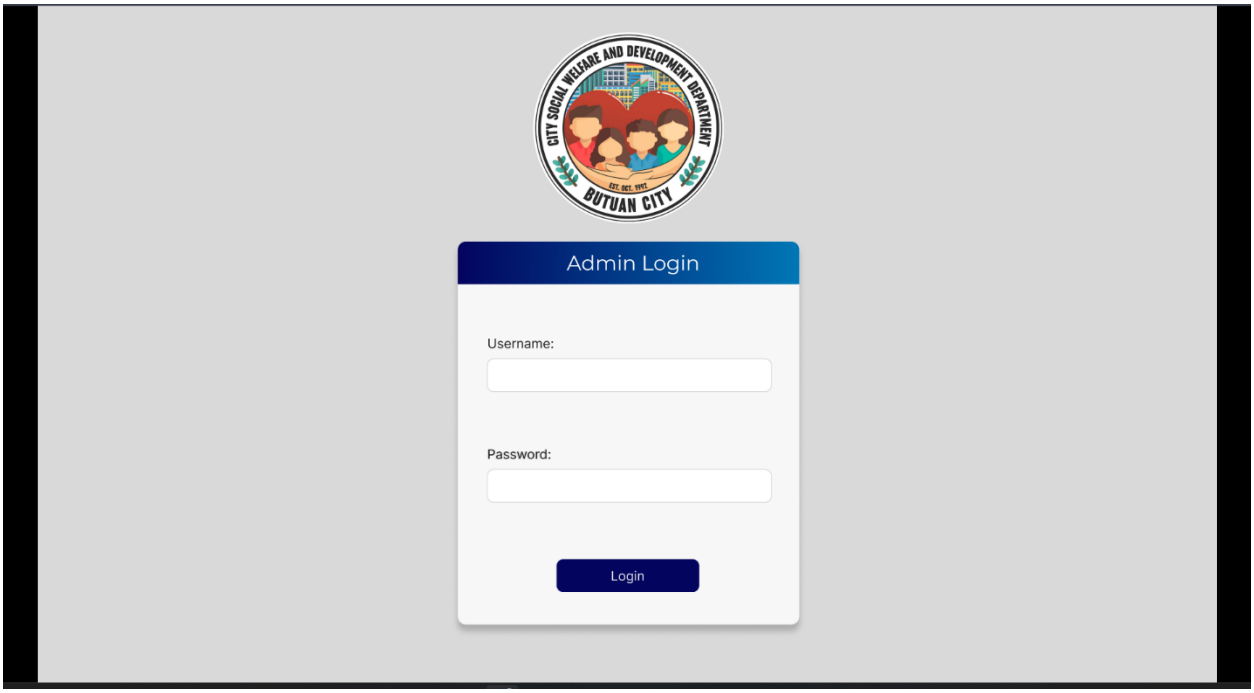


Figure 17. Admin login page

Figure 17 shows the Admin Login Interface for the City Social Welfare and Development Department of Butuan City. This is a safe way for allowed administrative staff to get in. The circular logo at the top of the screen, which has four stylish people and the department's name on it, reinforces the official system's identity and purpose to serve the community. The blogging form has two spaces for Username and Password, clearly labeled "Admin Login" under the logo. This makes sure that only verified users can access system-sensitive functions. After entering the user's credentials, clicking the blue "Login" button starts the verification process. This usually includes checking the user's input against a secure database and starting a session for them. The simple layout and minimalist design will put clarity and usefulness first, making it easy for administrators to log in quickly while still having full control over who

can access and protect data. This interface is a very important part of protecting the CCIMS platform's integrity and making sure that only allowed people do administrative tasks.

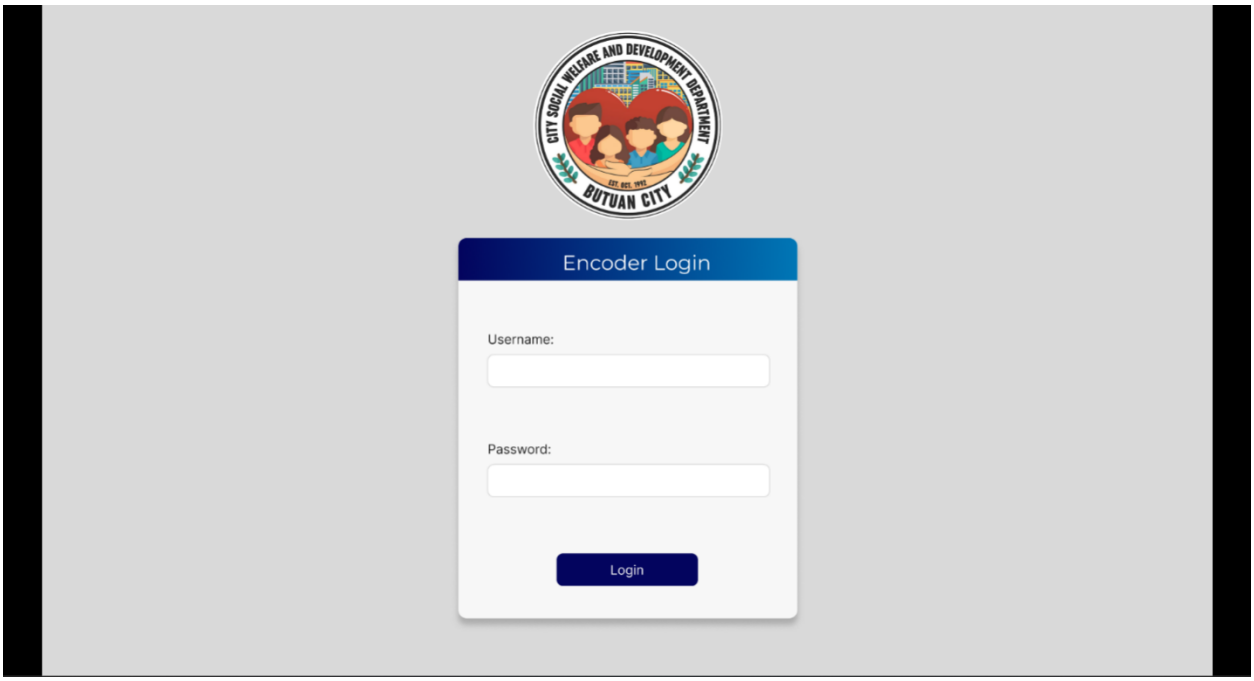


Figure 18. Encoder login page

Figure 18 The Encoder Login interface is designed for staff role in entering and managing client data within the CSWDD. At the top, the circular logo featuring three figures holding the hand symbolizes the unity of unity and community, which aligns with the department's mission to support the vulnerable populations. Under the logo, the form titled Encoder Login includes fields for the username and password, allowing encoders to safely access the system. When credentials are submitted via the Dark Blue Login button, the system is likely to confirm the user against a protected database and provides access to coding -such as registering new clients, updating service records, and program data management. Clean layout and neutral background emphasize professionalism and availability, ensuring that data entry staff can perform their activities efficiently while maintaining system integrity and confidentiality.

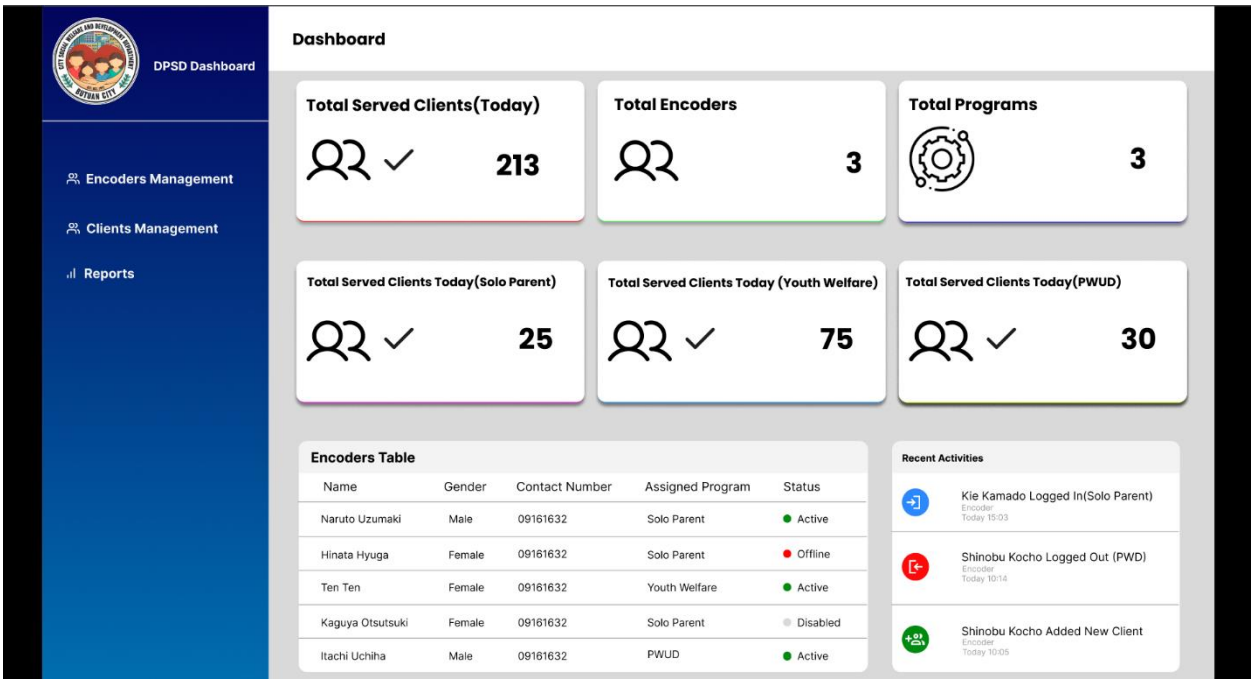


Figure 19. Admin dashboard

In Figure 19 The DPSD dashboard you show is structural and functionally aligned with the dashboards used by the PSSD and RPSD, even though each division serves different client sectors. The shared design ensures the same with the departments, allowing administrators and encoders to navigate and manage data easily regardless of their designated program.

Each dashboard shows real-time statistics such as total clients served today, total encoders, and number of specific client programs (e.g., solo parents, young people, PWD), which helps monitor service delivery performance. The Encoders table lists the details of staff such as name, gender, contact number, designated program, and status - easy to monitor who is active and where they are deployed. Recent panel activities have been logging user actions, strengthening responsibility and transparency.

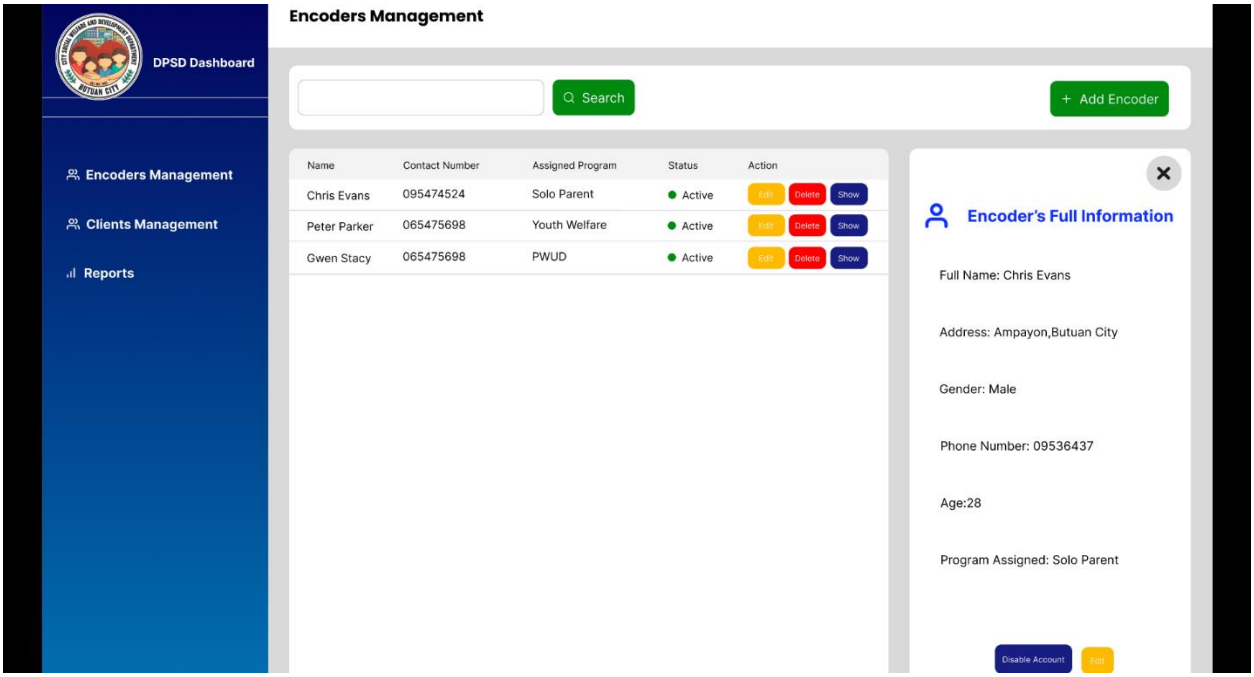


Figure 20. Encoders Management (Admin)

In Figure 20 The Encoder management interface you have shown from the DPSO Dashboard has been mashed with PSSD and RPSD, which ensures the consistency of how to handle encoder data in all program areas.

This shared layout includes a sidebar for navigation, a list of central encoder table details (name, contact number, designated program, status), and action buttons for editing, viewing, or removing records. On the right, a detailed panel shows full encoder information, including address, age, gender, and program assignment, along with a disabled account button for control. Having a search bar and adding an encoder button is a user management, making it easy to find or aboard staff.

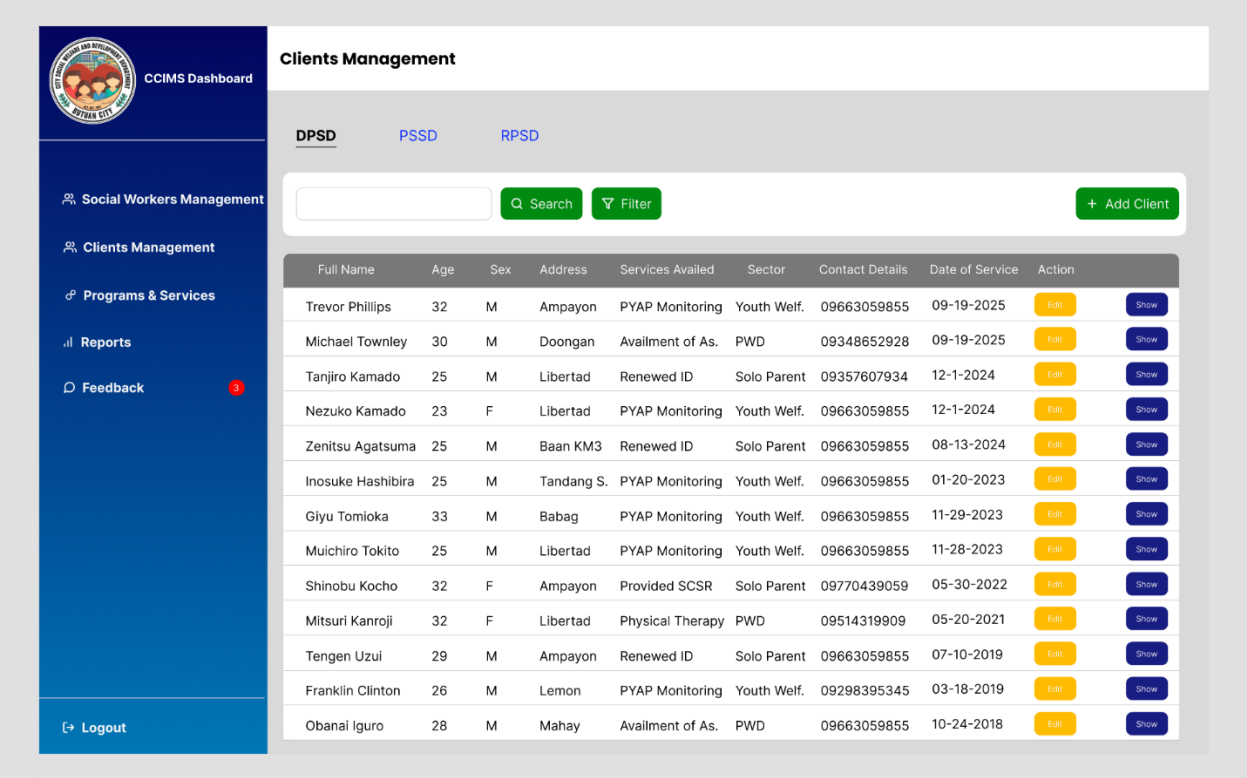


Figure 21. Clients Management (Admin)

In Figure 21 The interface of client management shows the part of a system of system architecture used in many divisions within the social welfare department. Although each division holds different areas of the program - such as senior citizens, people with disabilities, solo parents, youth welfare, and various protection services the interface design and basic functioning remain the same. Users can navigate through the attached views for each client category, with each tab that presents a structured table that includes important fields such as full name, age, gender, address, service available, sector, contact details, service date, and encoder. This standard layout ensures accurate documentation of client contacts and supports good service monitoring. By maintaining a constant workflow work in all divisions, the system simplifies user training, strengthens data integrity, and provides to the measured expansion. If the management of clients under any program or division, the experience remains intuitive and cohesive - it is easier to produce reports, monitor performance, and effective services.

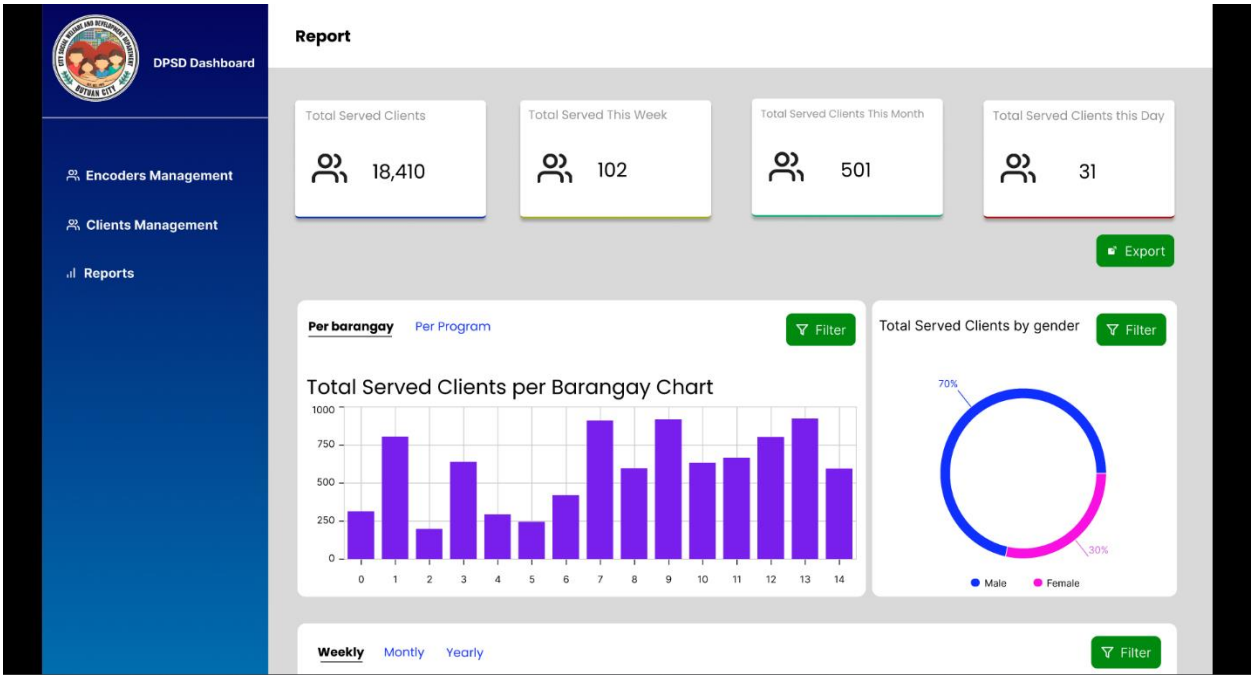


Figure 22. Statistical Report (Admin)

In Figure 22 shows the report provides a visual and statistical overall client data on all divisions, including PSSD, RPSD, DPSD, and others. It features summary panels showing the total clients served by the sun, weeks, and months, along with charts that destroy the village and gender service distribution. These visualizations help administrators quickly assess performance trends, identify service gaps, and make data -driven decisions. Incorporating filter options (e.g., each barangay, each program) and export operating supports the targeted review and reporting for internal reviews or external presentations. Despite the differences of the program's focus throughout the divisions, the module reports maintain a consistent layout and logic, making it easy for users to interpret data regardless of the sector they are studying. This one reporting structure reinforces the system scalability and ensures that all divisions can monitor the impact and performance using the same tools.

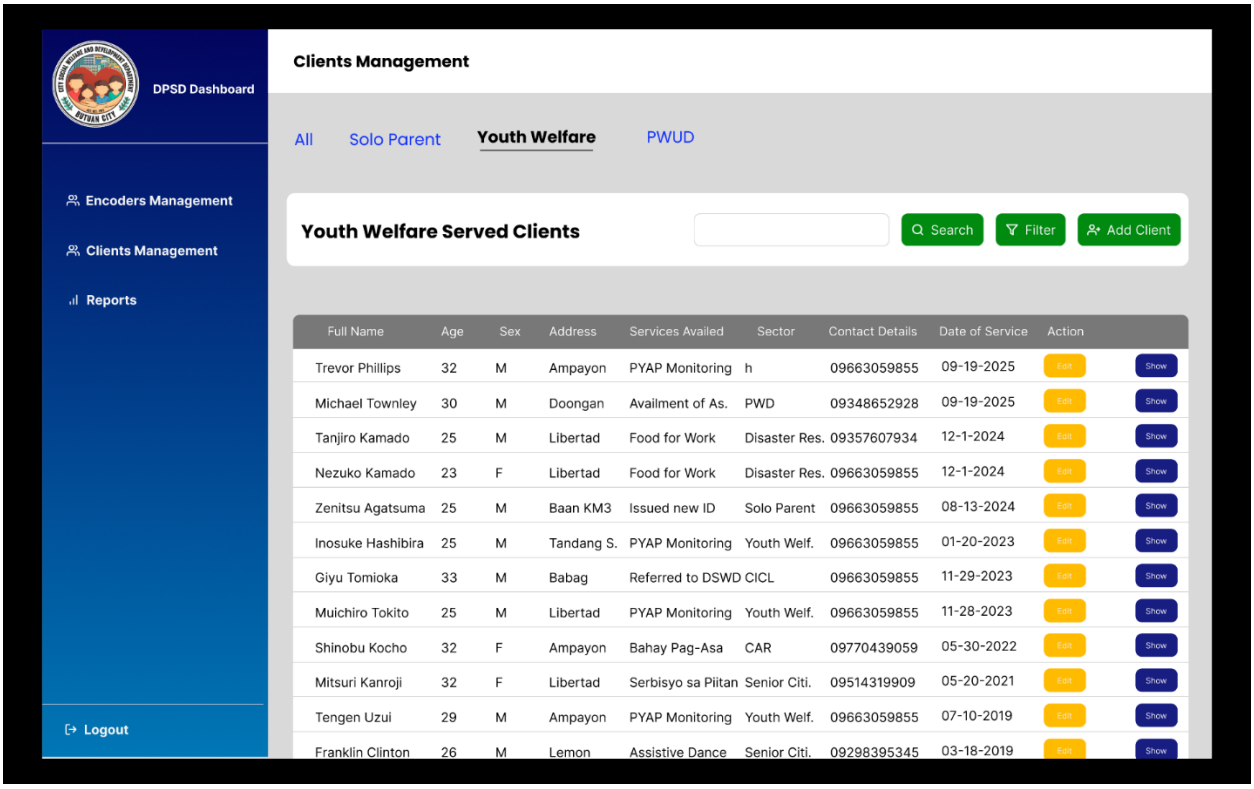


Figure 23. Program Served Clients

In Figure 23 The interface shows a filter table titled Youth of Youth served to clients, showing only the notes associated with the Encoder's assignment. Each row includes detailed client information such as full name, age, gender, address, service, sector, service details, service date, and contact number. It allows encoders to verify entries, monitor service delivery, and ensure accurate documentation for their designated sector.

While the layout and functionality are consistent with divisions such as the PSSD and RPSD, the Encoder's POV is a specific program. This means that encoders only see and manage the data tied to their designated program (e.g., solo parents, PWD, CIU, VAWC), maintenance of data privacy and operational focus. This paper-based filtering improves liability and ensures that each encoder works well within their scope, while still benefiting the unified system design.



Figure 24. Membership Information Sheet

In Figure 24 shows the Membership Information Sheet is a standard digital form of use used under the City Social Welfare and Development Department (CSWDD) under youth welfare program. Access to the DPSD dashboard management module, this form is specifically designed for encoders assigned to the youth welfare sector. It serves as an official registration tool for youth beneficiaries, especially those associated with Pag-asa Youth Association of the Philippines (PYAP).

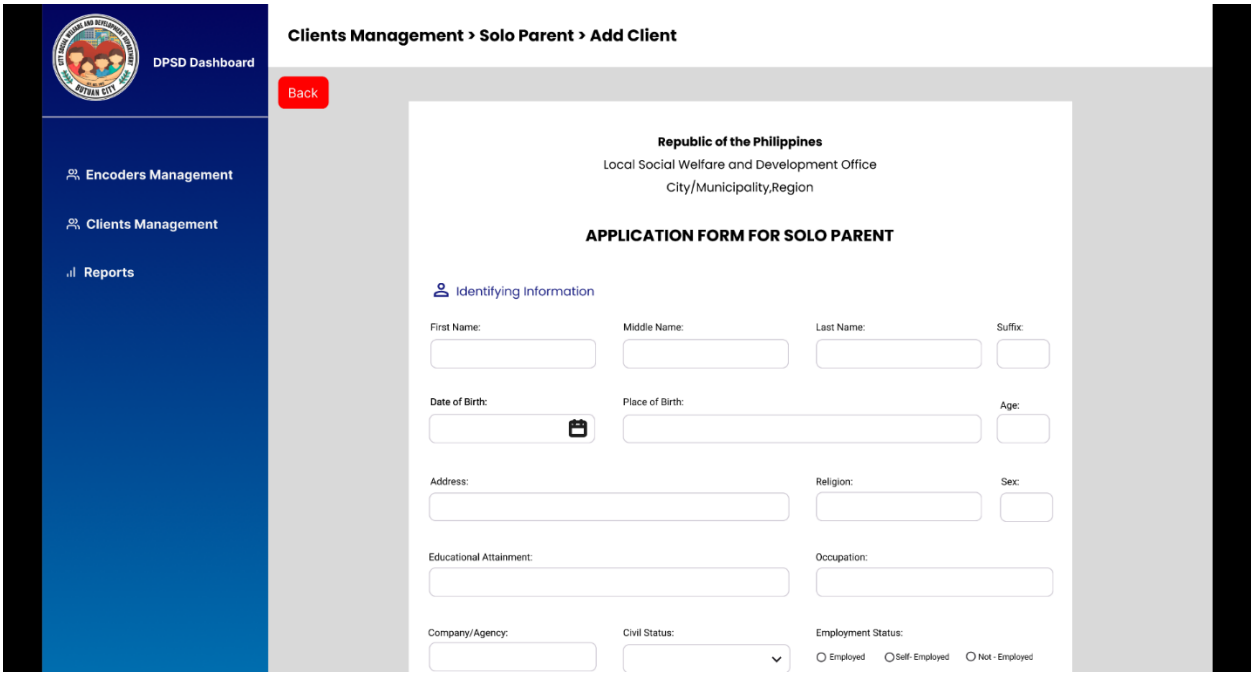


Figure 25. Application Form for Solo Parent

In Figure 25 show the application form for solo parent is a standard digital form of use used under the solo parent program of the Development and Promotive Services Division (DPSD), implemented through the DPSD dashboard client management module. This form is accessible by encoders assigned to the solo parent sector and serves as the official profiling tool for individuals seeking support and recognition under Republic Act No. 8972, also known as the Solo Parents Welfare Act.

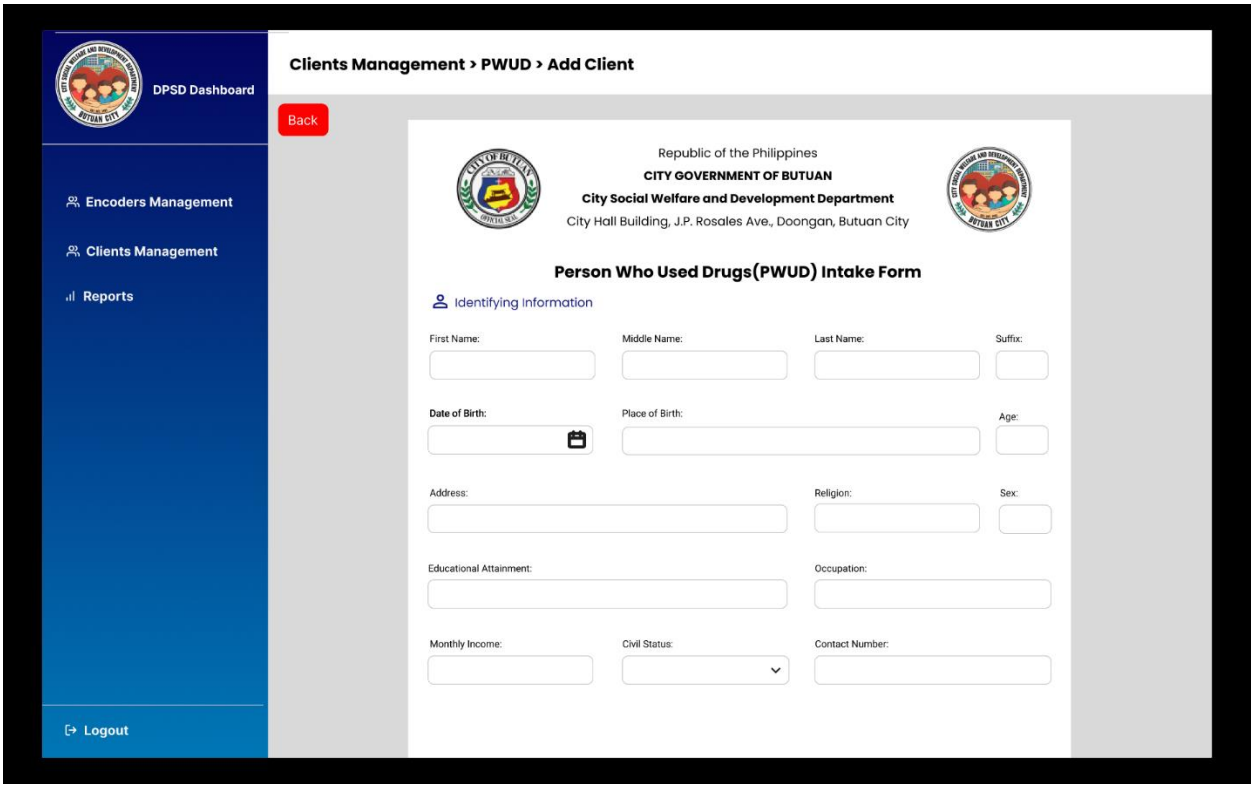


Figure 26. PWUD Intake Form

In Figure 26 shows the form of use of PWUD is a standard digital registration tool used by the City Social Welfare and Development Department (CSWDD) to profit individuals identified as drug users (PWUD). This form is accessible through the department's dashboard management client module and is usually filled with encoders assigned to the PWUD program. It serves as a preliminary step in documenting client information for rehabilitation, intervention, and support services.

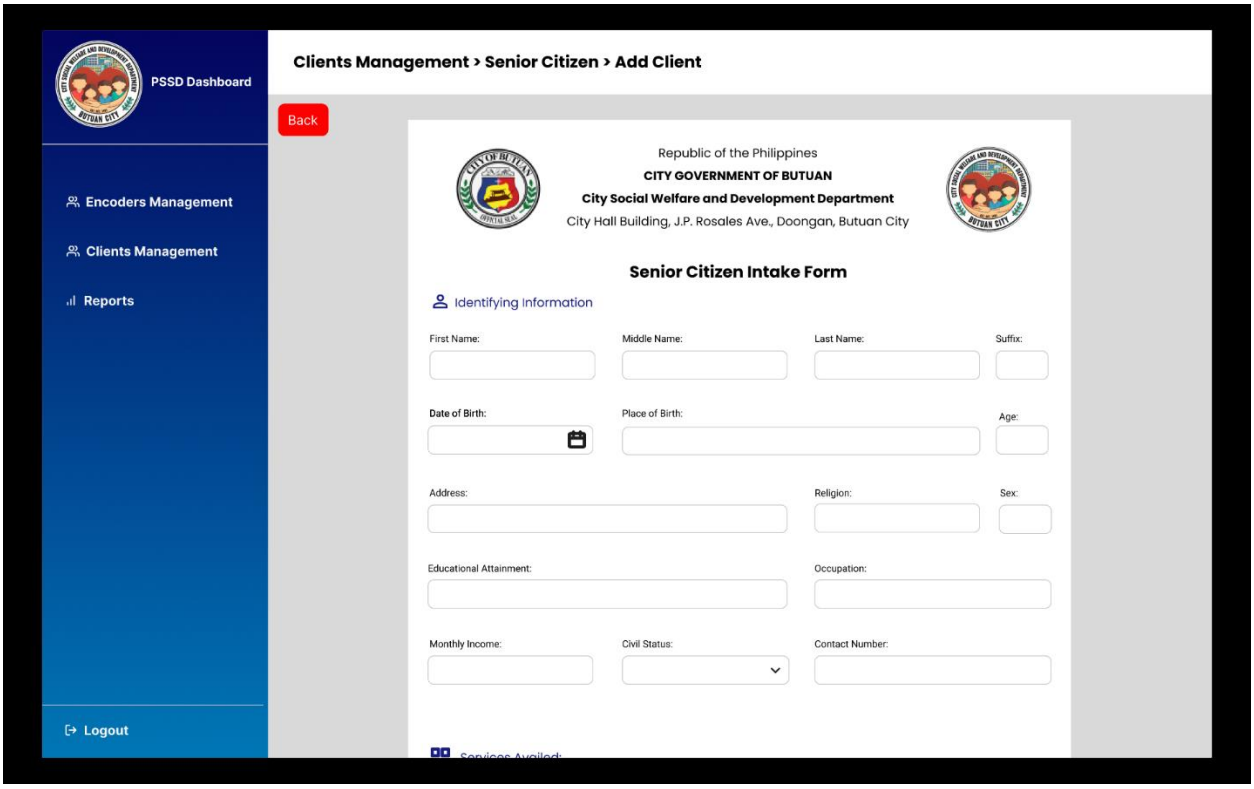


Figure 27. Senior Citizen Intake Form

In Figure 27 shows the Senior Citizen Intake Form is a standard digital registration tool used under the OSCA/PDAO Division of the City Social Welfare and Development Department (CSWDD), specifically designed to profile old clients seeking support for society. This form is accessible through the PSSD dashboard management module and completed by encoders assigned to the Senior Citizen Program.

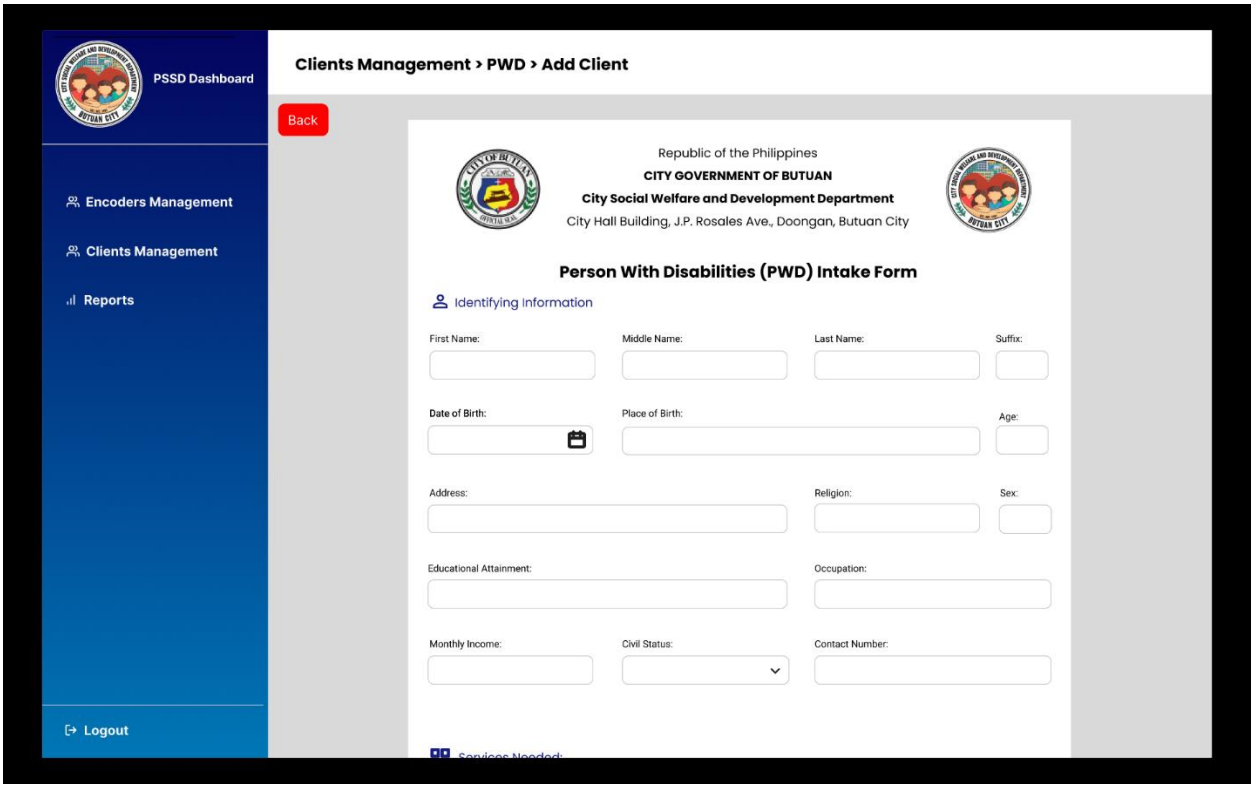


Figure 28. PWD Intake Form

In Figure 28 shows the form of PWD use is a digital registration tool used under the disabled (PWD) program, managed by the PSSD Division of the City Social Welfare and Development Department (CSWDD) in Butuan City. This form is accessible through the PSSD dashboard management module and completed by encoders assigned to the PWD sector. It serves as an official profiling mechanism for individuals seeking support, services, or recognition under disability -related programs.

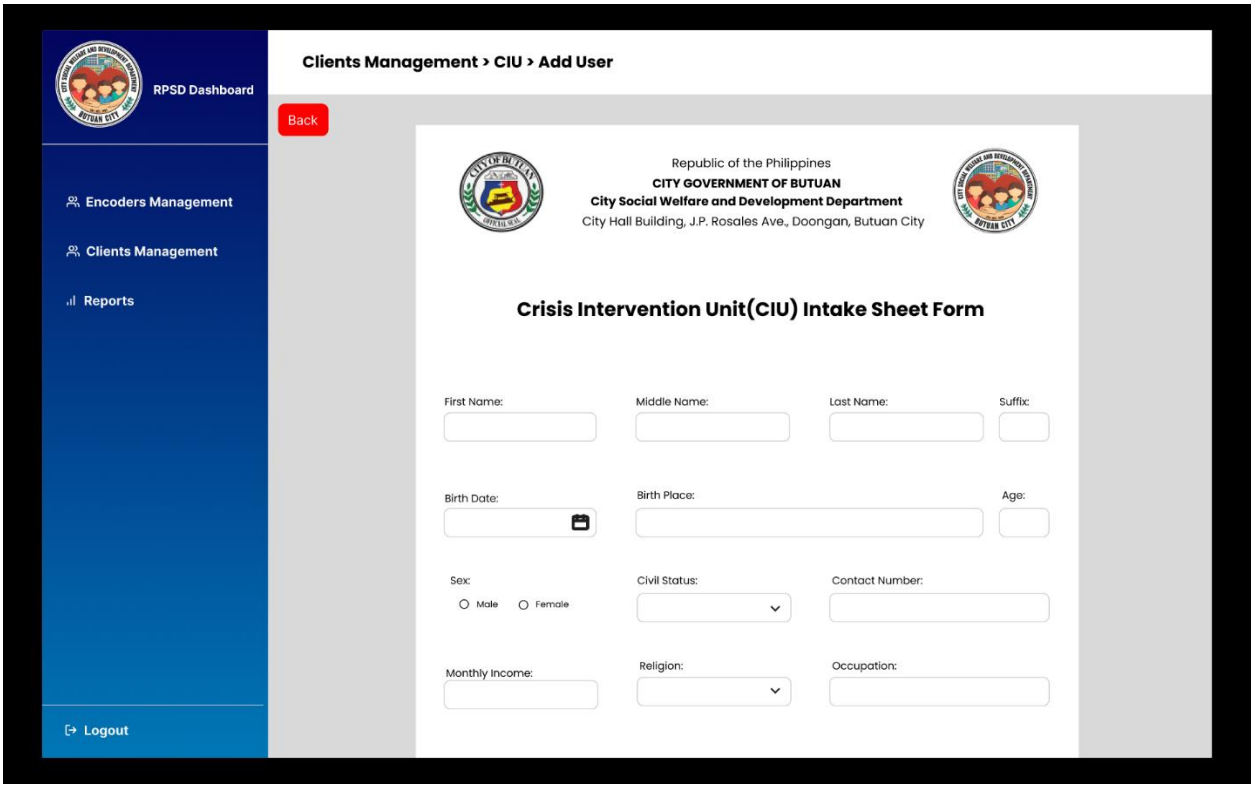


Figure 29. CIU Intake Sheet Form

In Figure 29 shows the form of CIU use is a digital registration tool used under the Crisis Intervention Unit (CIU) program, managed by the RPSD Division of the City Social Welfare and Development Department (CSWDD). This form is accessed through the RPSD dashboard management module and completed by encoders assigned to CIU-associated services. It serves as an official use mechanism for individuals seeking urgent help due to crisis situations such as medical emergency, family death, or financial anxiety.

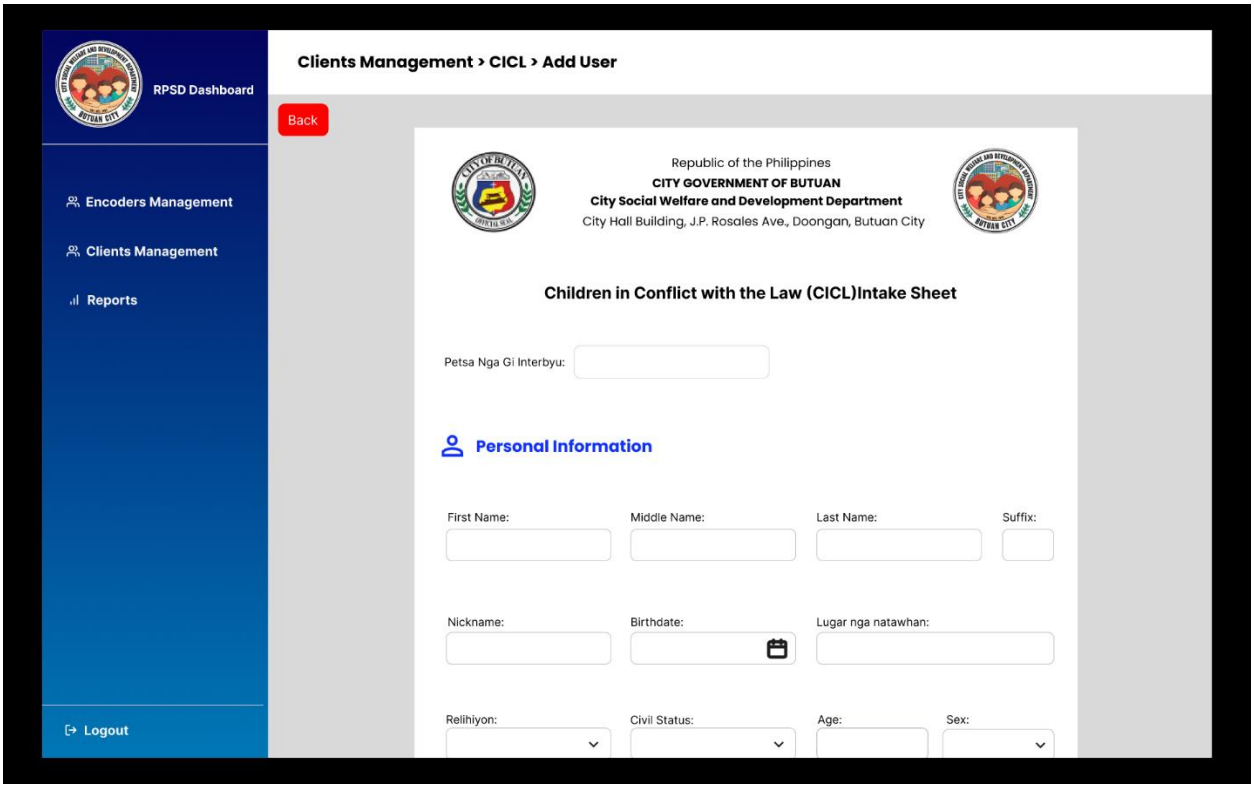


Figure 30. Intake Sheet of CICL and CAR

In Figure 30 shows form of the Children In Conflict with the Law program (CICL) and Children at Risk program (CAR) share a one -sized use structure because they both fall under the broader framework of protection services managed by the RPSD Division of the City Social Welfare and Development Department (CSWDD).

By using a unified use sheet, the system simplifies onboarding, supports cross-program coordination, and ensures that both CICL and CAR cases are recorded with equally strict and sensitive.

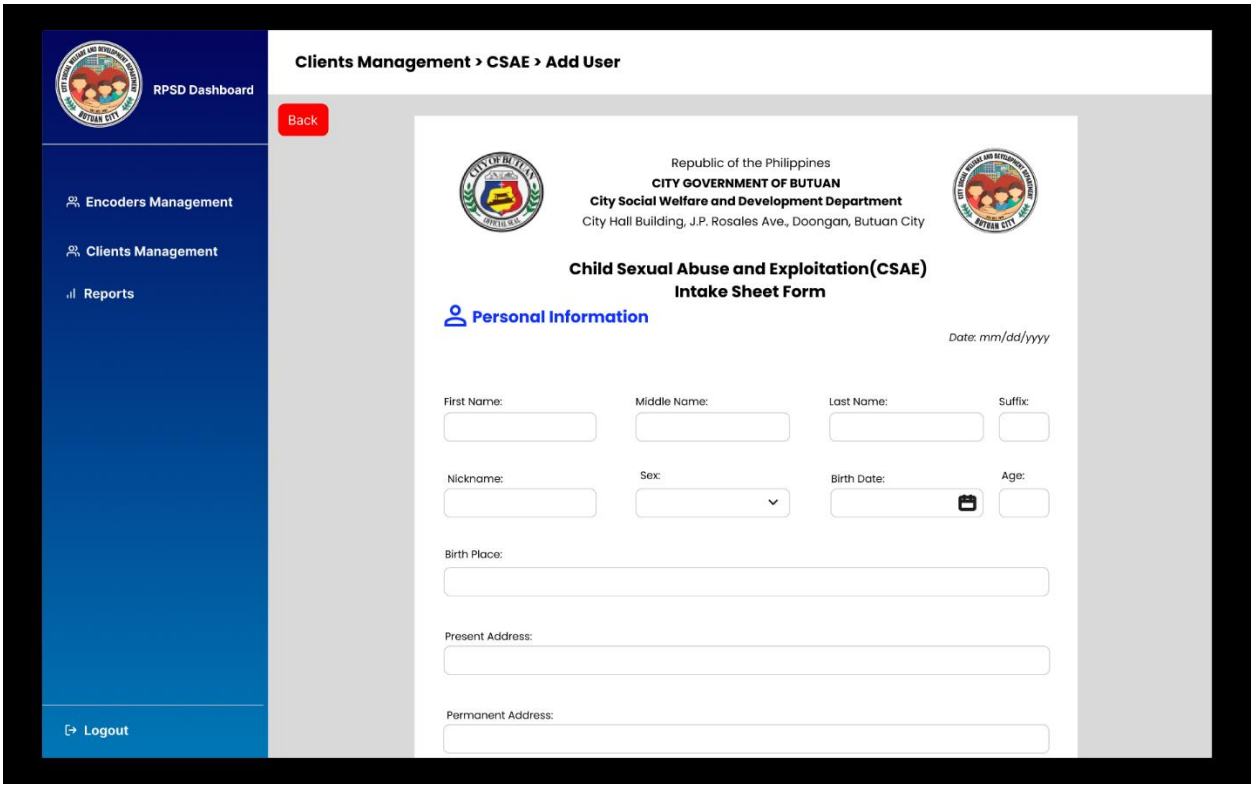


Figure 31. CSAE Intake Sheet Form

In Figure 31 shows the form of CSAE use is a digital registration tool used under children who are sexually abused and exploited (CSAE) program, managed by the RPSD Division of the City Social Welfare and Development Department (CSWDD). This form is accessible through the RPSD dashboard management module and completed by encoders assigned to child protection services. It serves as an official profiling mechanism for minor victims of sexual abuse, exploitation, or related offenses.

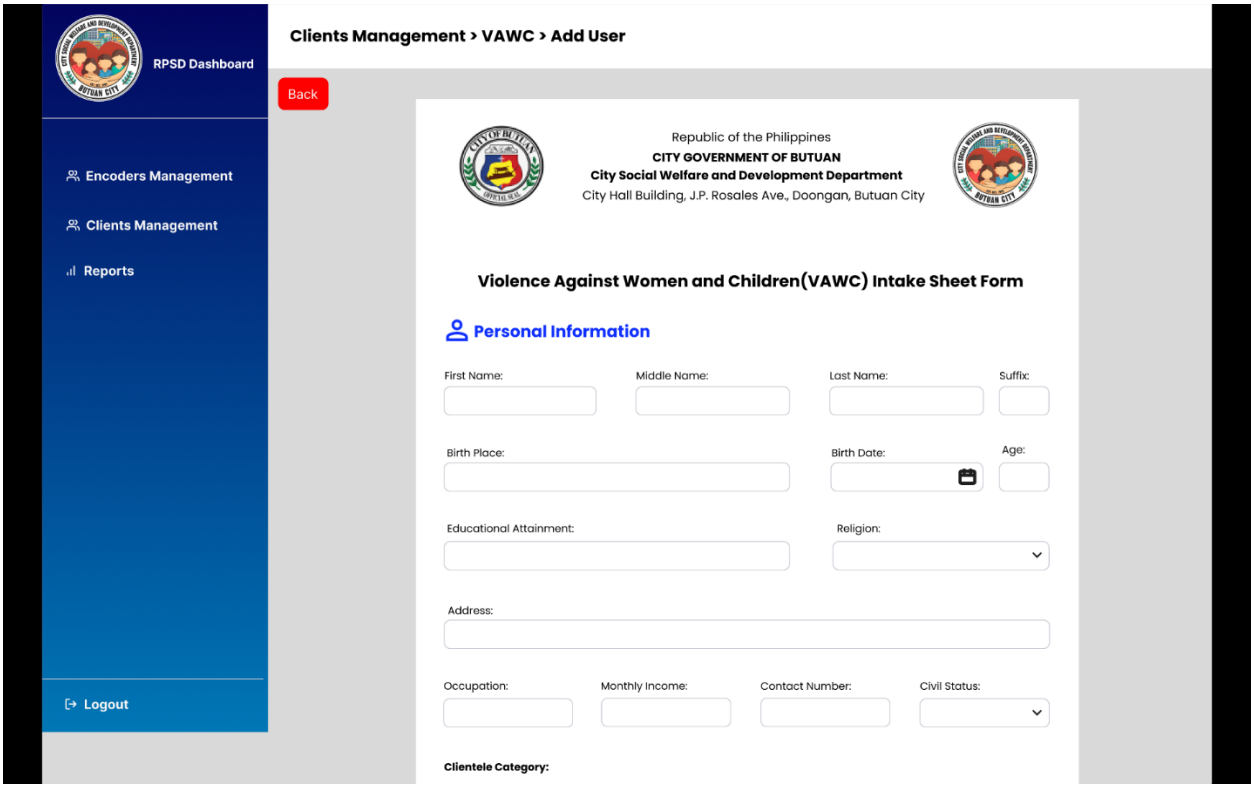


Figure 32. VAWC Intake Sheet Form

In Figure 32 shows the form of VAWC use is a digital registration tool used under the Women and Children (VAWC) program, managed by the RPSD Division of the City Social Welfare and Development Department (CSWDD). This form is accessed through the RPSD dashboard management module and completed by encoders assigned to the services associated with VAWC. It serves as an official profiling mechanism for individuals - mainly women and children - who are victims of abuse, exploitation, or violence.

Table 1

Software Platforms, Development Environment and Tools

Components	Specification	Usage
Internet Browser	Any (Chrome, Edge, Firefox, etc.)	Users can access and view information from the internet with a web browser. After a successful login, the CCIMS interface loads, allowing users to access its functionalities.
Front-end	HTML	HTML specifies the structural elements of the CCIMS Workplace System's interface, including forms, text, and other critical components.
	Tailwind CSS	CSS framework that allows developers to design custom, modern-looking interfaces quickly.
	JavaScript	To make the website more interactive and provide users with more functionality, JavaScript handles all client-side functionality.
Backend	MySQL (Database)	Handles structured data storage and management. Through phpMyAdmin, developers can create tables, run queries, and maintain database integrity for the System.
	PHP	PHP is used on the server side to manage interactions between user input and the database, ensuring smooth data flow.
Server	Apache	Using the XAMPP control panel, developers can manage the behavior of its components, ensuring

		smooth, easy access to local host servers such as Apache and MySQL.

Table 2

Hardware Requirements

Components	Specification	Usage
Device	RAM (16 GB)	This is like the computer's short-term memory. It helps the computer run many things at once and makes apps and files open faster.
	Processor (Intel Core i5)	The main component of the computer—the processor, or CPU—is responsible for executing commands from the website. It also makes data processing easier.
	SSD (128 GB)	This is where the computer saves programs, reports, and files. It works faster than old hard drives, so you can open and save things quickly.
	Flash Storage (8 GB)	These storage solutions ensure quick retrieval and secure long-term data handling.
Printer	Any Ink Jet Printer Units	Inkjet printers produce clear, high-quality output used to print compliance documentation, data summaries, and audit reports. Since these printers are reasonably priced and multipurpose, they can be used for routine audit management printing,

		including creating formal documents for internal use or sending them to stakeholders.

Research Ethical Standards

A. Protection of Intellectual Property Rights (IPR)

The researchers ensured the protection of intellectual property rights throughout the development of the Centralized Client Information Management System (CCIMS). This study complies with Republic Act No. 8293, also known as the Intellectual Property Code of the Philippines, which protects original works, including computer programs, databases, and system documentation.

All system source codes, system designs, database structures, and related documentation were originally developed by the researchers. Ownership of the developed system is intended for academic purposes and official use by the Butuan City Social Welfare and Development Department (BCSWDD), subject to institutional policies and agreements. The researchers acknowledge that any further modification, deployment, or commercialization of the system must be conducted with proper authorization from the concerned institution and stakeholders.

The system was developed using open-source technologies such as PHP, MySQL, JavaScript, Apache, and Tailwind CSS. These technologies are used in compliance with their respective licenses, including the MIT License, GNU General Public License (GPL), and Apache License. Proper attribution is given to all third-party tools and libraries utilized in the development of the system. No copyrighted software or proprietary source code was used without authorization.

By observing these ethical standards, the researchers ensured respect for intellectual property rights, legal compliance, and responsible software development practices.

B. Informed Consent

Prior to participation in the system evaluation, all administrators, encoders, and technical staff from the City Social Welfare and Development Department (CSWDD) were provided with a written informed consent form. The consent form clearly explained the purpose of the study, its scope, the procedures involved, and the expected duration of participation.

Participants were informed of any possible risks, which were minimal and limited to time spent during system testing, as well as the anticipated benefits, such as contributing to the improvement of service delivery and system usability. The consent form also stated that participation in the study was entirely voluntary and that participants had the right to decline or withdraw from the study at any time without any form of penalty, consequence, or effect on their professional standing.

Written consent was obtained from all participants before data collection and system testing began, confirming that they fully understood the study and agreed to participate willingly. These measures ensured that ethical standards regarding participant autonomy, transparency, and voluntary participation were strictly observed throughout the research process.

C. Data Privacy and Confidentiality

This study strictly complies with Republic Act No. 10173, also known as the Data Privacy Act of 2012, which mandates the protection of personal information collected, stored, and processed in information systems.

During the system testing and evaluation phase, no actual client records or personally identifiable information were used. All data utilized in the system were anonymized or fictitious, ensuring that no real individual could be identified. Any sample data entered into the system were generated solely for testing and demonstration purposes.

To ensure data security, the Centralized Client Information Management System (CCIMS) implements role-based access control (RBAC), secure user authentication, and password-protected accounts, limiting system access only to authorized users such as the Super Admin, Admin, and Encoder. The database is stored in a secured MySQL server environment hosted on a local server, protected through access restrictions and administrative controls to prevent unauthorized access.

All research-related data were accessible only to the researchers and authorized system administrators. No data whether anonymized or aggregated were shared with external parties without prior written authorization from the concerned institution.

In compliance with ethical and legal standards, all temporary data used during testing were retained only for the duration of the study and were securely deleted after the completion of system evaluation and documentation. These measures ensured confidentiality, integrity, and lawful handling of data throughout the research process.

D.Voluntary Participation and Freedom to Withdraw

Participation in this study was entirely voluntary. All administrators, encoders, and technical staff from the City Social Welfare and Development Department (CSWDD) who participated in the usability and functionality testing were clearly informed of their right to refuse participation or withdraw from the study at any stage without facing any form of penalty, disadvantage, or negative consequence.

The researchers explicitly ensured that no coercion, pressure, inducement, or undue influence was used to encourage participation. Participants' decisions to join, decline, or withdraw from the study had no effect on their employment status, professional responsibilities, or relationship with the institution.

Since the study involved only system evaluation and did not include animals, biological materials, or clinical experimentation, no animal-related ethical concerns were involved.

E. Minimization of Harm and Risk Management

This study was conducted with the primary objective of causing no harm to participants and involved minimal risk only. Participation was limited to system usage, usability testing, survey responses, and feedback collection related to the evaluation of the Centralized Client Information Management System (CCIMS). No physical, psychological, emotional, or social harm was anticipated or experienced by any participant during the conduct of the study.

To manage potential risks, the researchers implemented appropriate risk management procedures, including conducting all testing in a controlled environment, using anonymized or fictitious data, and restricting system access to authorized users only. Participants were encouraged to raise any concerns during the testing process, and the researchers immediately provided clarification, technical assistance, or discontinued participation if necessary.

In compliance with ethical research standards, appropriate safeguards were applied to prevent data exposure, system misuse, or participant discomfort. These safeguards ensured that participants were protected throughout the study and that any potential risk was promptly addressed.

Furthermore, the study adheres to the ethical principles of the 3Rs (Replacement, Reduction, and Refinement). Since the research did not involve animals, biological materials, or experimental testing, replacement was applied by utilizing human-centered system evaluation methods, reduction was ensured by limiting participation to essential respondents only, and refinement was observed by designing procedures that minimized time, effort, and inconvenience to participants.

F. Beneficence and Contribution to Knowledge

This study adheres to the ethical principle of beneficence by ensuring that the outcomes of the research provide clear benefits to the institution, participants, and the wider community. The development of the Centralized Client Information Management System (CCIMS) directly benefits the City Social Welfare and Development Department (CSWDD) by improving the efficiency, accuracy, and organization of client data, thereby supporting more effective and timely service delivery.

Participants and staff involved in the system evaluation benefited through exposure to an improved digital workflow that reduces manual processing, minimizes errors, and enhances data accessibility. In turn, the community served by CSWDD benefits from more transparent, responsive, and data-driven social welfare services.

From an academic perspective, the study contributes to the body of knowledge in information systems development by providing documented design processes, implementation strategies, and evaluation results related to centralized government information systems. The findings, system documentation, and research outputs will be made available to the academic community, participants, and institutional partners upon request, supporting future research, system improvement, and replication efforts.

Through these contributions, the study ensures that its benefits extend beyond system development and provide meaningful academic and practical value.

G. Justice and Fair Participant Selection

This study adhered to the ethical principle of justice by ensuring fair, unbiased, and appropriate participant selection. Participants were selected based on clear inclusion and exclusion criteria aligned with the objectives of the research.

Inclusion criteria consisted of authorized administrators, encoders, and technical staff of the City Social Welfare and Development Department (CSWDD) who were directly involved in client data management, system operation, or system evaluation. These participants were selected to ensure relevance to the system's functionality and to maintain data security.

Exclusion criteria included individuals who were not involved in system operations, had no official role in client data handling, or were not authorized by the institution to access the system. Members of the general public and vulnerable populations were excluded to prevent unnecessary risk and ensure ethical compliance.

The selection process was conducted without bias or discrimination based on gender, age, socioeconomic status, religion, or personal background. All eligible participants were given equal opportunity to participate, ensuring fairness and appropriate representation within the scope of the study.

By applying these criteria, the study ensured that participant selection was just, equitable, and ethically appropriate, while safeguarding both institutional data and participant welfare.

H. Data Integrity and Accuracy

Data integrity and accuracy were maintained throughout the study through the use of standardized data-handling and system-testing procedures. The system evaluation was guided by the ISO/IEC 25010 Software Quality Model, which served as the benchmark for assessing system characteristics such as functionality, performance efficiency, reliability, usability, and security.

To ensure accuracy and reliability, system testing followed a structured validation process, including functionality testing, usability evaluation, and verification of system outputs against expected results. All data entries, system responses, and computed outputs were reviewed and cross-checked by the researchers to minimize errors and inconsistencies. Any detected issues were corrected and retested to confirm system accuracy.

Measures were also taken to ensure data consistency, including the use of predefined input formats, controlled testing conditions, and uniform evaluation criteria across all participants. Potential limitations and sources of bias, such as participant familiarity with digital systems or limited testing duration, were identified, documented, and acknowledged in the Results and Discussion sections of the study.

By adhering to recognized international standards and validation procedures, the researchers ensured that the data collected and analyzed were accurate, reliable, and ethically sound.

I. Transparency and Honesty in Reporting

The researchers are committed to full transparency and honesty in the reporting of all research activities and findings. All data, observations, and results presented in this study were reported accurately, completely, and truthfully, with no fabrication, falsification, manipulation, or selective omission of data at any stage of the research process.

The research report includes all relevant methodologies, system evaluation results, limitations, and findings, regardless of whether the outcomes aligned with the researchers' initial expectations. This ensures the completeness and reliability of the research report and allows for proper interpretation and replication of the study.

All referenced literature, software tools, frameworks, and external resources were properly cited and acknowledged in accordance with accepted academic and ethical standards. The researchers also declare that there were no conflicts of interest, financial or otherwise, that could have influenced the conduct, analysis, or reporting of the study.

Through these practices, the study upholds the principles of research integrity, accountability, and ethical scholarship.

J.Use of Patented or Copyrighted Materials

The development of the Centralized Client Information Management System (CCIMS) utilized open-source software technologies, including PHP, MySQL, HTML, CSS, and JavaScript, which are legally available for academic and development use under their respective open-source licenses. These technologies were used in full compliance with their licensing terms, including licenses such as the MIT License, GNU General Public License (GPL), and other applicable open-source usage agreements.

No patented, proprietary, or restricted software tools were used in the development of the system without proper authorization or permission from the rights holders. All third-party libraries, frameworks, and external code snippets were employed strictly within the scope allowed by their licenses.

All copyrighted materials, including academic literature, documentation, images, and external references, were used solely for educational and research purposes and were properly cited and acknowledged in accordance with intellectual property laws and academic standards. The researchers ensured that the use of all software and materials respected software usage rights, attribution requirements, and ethical research practice.

K. Ethical Consideration for Animals and Human Trials

This research did not involve any animal or biological testing. Human involvement was limited to authorized CSWDD staff who participated in usability evaluation and feedback sessions. All participants were briefed on the potential benefits and minimal risks before engagement. Ethical guidelines for human

interaction were observed throughout the study to protect participants' rights and well-being.This research did not involve any animal experimentation, biological testing, or clinical trials.

This research did not involve any animal experimentation, biological testing, or clinical trials. Human involvement was limited to authorized administrators, encoders, and technical staff of the City Social Welfare and Development Department (CSWDD) who participated solely in system usability evaluation and feedback activities.

Participation in the study was entirely voluntary, and all participants were fully informed of their right to decline or withdraw from the study at any time without any penalty or negative consequences. Prior to participation, written informed consent was obtained, and participants were briefed on the purpose of the study, the procedures involved, and the minimal risks and expected benefits.

In compliance with Republic Act No. 10173 (Data Privacy Act of 2012), no personally identifiable information or real client data were collected or processed during the study. All test data used in system evaluation were anonymized or fictitious, and access to the system was restricted to authorized personnel only. These measures ensured the confidentiality, privacy, and protection of participant information.

The study was conducted in accordance with established ethical guidelines for human-centered research and was subject to review and approval by the institution’s research ethics committee or designated academic authority prior to implementation. Throughout the research process, appropriate ethical safeguards were observed to protect the rights, safety, and well-being of all participants.

L. Responsible Use of Artificial Intelligence (AI) and Related Technologies

This study acknowledges the responsible and ethical use of artificial intelligence (AI)–assisted tools during the research and documentation process. AI-based applications such as ChatGPT and Grammarly were utilized solely as support tools to assist in grammar checking, language refinement, and clarification of ideas during the writing process.

The use of these tools did not replace the researchers’ original work, analysis, system design, programming, or decision-making. All research content, system development, data interpretation, and conclusions were independently conducted by the researchers. AI tools were not used to fabricate data, manipulate results, or generate system outputs.

The researchers ensured that the use of AI-assisted technologies complied with academic integrity policies and ethical research standards. Proper judgment was exercised to verify the accuracy, relevance, and appropriateness of all content included in the study. The researchers remain fully accountable for the originality, validity, and integrity of the research work.

By disclosing the use of AI tools, this study promotes transparency, accountability, and responsible innovation in the application of emerging technologies in academic research.

M. Ethical Clearance and Institutional Approval

This research was conducted in accordance with the ethical guidelines and academic research policies of Saint Michael College of Caraga. Prior to the implementation of the study, the research proposal, system design, and data-handling procedures were reviewed and approved by the institution's designated research ethics committee, research coordinator, or authorized academic authority.

Formal permission to conduct the study was also obtained from the City Social Welfare and Development Department (CSWDD), ensuring that the system evaluation, data usage, and participant involvement were conducted with institutional awareness and authorization.

All research activities complied with approved protocols, ethical standards, and legal requirements, including the Data Privacy Act of 2012 (RA 10173) and applicable institutional policies. Any modifications to the research procedures were coordinated with and approved by the appropriate academic authorities prior to implementation.

Through this ethical clearance and institutional approval, the researchers ensured that the study was conducted responsibly, transparently, and in full compliance with ethical research standards.

System Evaluation

The evaluation of the Centralized Client Information Management System was conducted using two approaches: the ISO 25010 quality model. This dual framework ensures that the System is not only functional, efficient, and usable but also that its AI component delivers accurate and reliable recommendations. *Data was analyzed and interpreted based on the following parameters:*

Table 3

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Functional

3	2.50 – 3.49	Functional
2	1.50 – 2.49	Moderately Functional
1	1.0 – 1.49	Poor Functional

Functional Suitability

Table 3 presents the criteria used to evaluate a product or System's functional suitability according to the ISO survey.

This measurement highlights the degree to which the System's modules—such as reports and AI analysis—align with user expectations and deliver the intended outcomes.

Table 4

Performance Efficiency

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Efficient
3	2.50 – 3.49	Efficient
2	1.50 – 2.49	Moderately Efficient
1	1.0 – 1.49	Poor Efficient

Table 4 outlines the scoring parameters used to assess the System's overall performance efficiency in executing its functions and tasks.

Table 5

Usability

Adjectival Rating	Scale Range (Mean)	Verbal Interpretation
4	3.50 – 4.0	Very Usable
3	2.50 – 3.49	Usable
2	1.50 – 2.49	Moderately Usable
1	1.0 – 1.49	Poor Usable

Table 5 presents the scoring parameters used to assess the System's usability.

Table 6

Respondents' Participated

Respondents	Number	Percentage
Super Admin	1	7%
Admin	3	21%
Encoder	10	72%
Total	14	100%

Table 6 presents the distribution of respondents who participated in the system evaluation of the Centralized Client Information Management System (CCIMS-BCSWDD). The respondents consisted of Super Admins, Division Admins, and Encoders from the City Social Welfare and Development Department (CSWDD). These individuals were selected based on their active roles in the use and management of client information. Their participation ensured that the evaluation results accurately reflected the perspectives of all key system users.

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